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# Contents

Intellectual Property Rights ..... 2
Legal Notice ..... 2
Modal verbs terminology ..... 2
Foreword ..... 6
1 Scope ..... 8
2 References ..... 8
3 Definitions of terms and abbreviations ..... 9
3.1 Terms ..... 9
3.2 Abbreviations ..... 9
4 General description ..... 10
5 Functional entities ..... 10
5.1 SEAL configuration management client (SCM-C) ..... 10
5.2 SEAL configuration management server (SCM-S) ..... 11
6 Configuration management procedures ..... 12
6.1 General ..... 12
6.2 On-network procedures ..... 12
6.2.1 General ..... 12
6.2.1.1 Authenticated identity in HTTP request ..... 12
6.2.1.2 Authenticated identity in CoAP request ..... 12
6.2.2 Common procedures ..... 12
6.2.2.1 Management of configuration update event subscription ..... 12
6.2.2.1.1 SIP based procedures ..... 12
6.2.2.1.2 HTTP based procedures ..... 14
6.2.2.1.3 CoAP based procedures ..... 15
6.2.2.2 Notifications ..... 16
6.2.2.2.1 SIP based procedures ..... 16
6.2.2.2.2 HTTP based procedures ..... 16
6.2.2.2.3 CoAP based procedures ..... 17
6.2.3 VAL UE configuration data ..... 17
6.2.3.1 SCM client HTTP procedure ..... 17
6.2.3.2 SCM server HTTP procedure ..... 17
6.2.3.3 SCM client CoAP procedure ..... 18
6.2.3.4 SCM server CoAP procedure ..... 18
6.2.4 VAL user profile data ..... 19
6.2.4.1 SCM client HTTP procedure ..... 19
6.2.4.2 SCM server HTTP procedure ..... 19
6.2.4.3 SCM client CoAP procedure ..... 19
6.2.4.4 SCM server CoAP procedure ..... 19
6.2.5 Update VAL user profile data ..... 20
6.2.5.1 SCM client HTTP procedure ..... 20
6.2.5.2 SCM server HTTP procedure ..... 20
6.2.5.3 SCM client CoAP procedure ..... 21
6.2.5.4 SCM server CoAP procedure ..... 21
6.3 Off-network procedures ..... 21
7 Coding ..... 22
7.1 VAL user profile document ..... 22
7.1.1 General ..... 22
7.1.2 Application unique ID ..... 22
7.1.3 Data structure ..... 22
7.1.4 XML Schema ..... 22
7.1.5 Semantics ..... 23
7.1.6 MIME type ..... 23

7.1.7 IANA registration template....23
7.2 VAL UE configuration document....25
7.2.1 General....25
7.2.2 Application unique ID....25
7.2.3 Data structure....25
7.2.4 XML schema....26
7.2.5 Semantics....27
7.2.6 MIME type....28
7.2.7 IANA registration template....28

**Annex A (normative): Parameters for different operations....31**

A.1 Creating configuration update event subscription....31
A.1.1 General....31
A.1.2 Client side parameters....31
A.1.3 Server side parameters....31

A.2 Retrieve VAL UE configuration data....31
A.2.1 Client side parameters....31

**Annex B (normative): Parameters for notifications....33**

B.1 General....33

B.2 Configuration update notification....33

**Annex C (normative): CoAP resource representation and encoding....34**

C.1 General....34
C.1.1 Resource URI structure....34
C.1.2 Use of cache....34
C.1.3 Error handling....34
C.1.4 Data types applicable to multiple resource representations....37
C.1.4.1 General....37
C.1.4.2 Referenced structured data types....37
C.1.4.3 Referenced simple data types and enumerations....37
C.1.4.4 Common structured data types....39
C.1.4.4.1 Type: ScheduledCommunicationTime....39
C.1.4.4.2 Type: ProblemDetails....39
C.1.4.4.3 Type: GeographicalCoordinates....39
C.1.4.4.4 Type: GeographicArea....40
C.1.4.4.5 Type: Point....40
C.1.4.4.6 Type: PointUncertaintyCircle....40
C.1.4.4.7 Type: PointUncertaintyEllipse....40
C.1.4.4.8 Type: Polygon....41
C.1.4.4.9 Type: PointAltitude....41
C.1.4.4.10 Type: PointAltitudeUncertainty....41
C.1.4.4.11 Type: EllipsoidArc....41
C.1.4.4.12 Type: UncertaintyEllipse....42
C.1.4.5 Common enumerations....42
C.1.4.5.1 Enumeration: SupportedGADShapes....42

C.2 Resource representation and APIs for VAL user profile....42
C.2.1 SU_UserProfile API....42
C.2.1.1 API URI....42
C.2.1.2 Resources....43
C.2.1.2.1 Overview....43
C.2.1.2.2 Resource: User Profiles....43
C.2.1.2.2.1 Description....43
C.2.1.2.2.2 Resource Definition....43
C.2.1.2.2.3 Resource Standard Methods....44
C.2.1.2.3 Resource: Individual User Profile....45
C.2.1.2.3.1 Description....45
C.2.1.2.3.2 Resource Definition....45

C.2.1.2.3.3 Resource Standard Methods ..... 45
C.2.1.3 Data Model ..... 46
C.2.1.3.1 General ..... 46
C.2.1.3.2 Structured data types ..... 47
C.2.1.3.2.1 Type: ProfileDoc ..... 47
C.2.1.3.2.2 Type: ProfileInfo ..... 47
C.2.1.3.2.3 Type: ProfileConfig ..... 47
C.2.1.3.2.4 Type: ValTargetUe ..... 47
C.2.1.3.3 Simple data types and enumerations ..... 48
C.2.1.3.3.1 Enumeration: ConfigType ..... 48
C.2.1.4 Error Handling ..... 48
C.2.1.5 CDDL Specification ..... 48
C.2.1.5.1 Introduction ..... 48
C.2.1.5.2 CDDL document ..... 48
C.2.1.6 Media Type ..... 49
C.2.1.7 Media Type registration for application/vnd.3gpp.seal-user-profile-info+cbor ..... 49
C.3 Resource representation and APIs for UE configuration ..... 50
C.3.1 SU_UeConfig API ..... 50
C.3.1.1 API URI ..... 50
C.3.1.2 Resources ..... 50
C.3.1.2.1 Overview ..... 50
C.3.1.2.2 Resource: UE Configurations ..... 51
C.3.1.2.2.1 Description ..... 51
C.3.1.2.2.2 Resource Definition ..... 51
C.3.1.2.2.3 Resource Standard Methods ..... 51
C.3.1.2.3 Resource: Individual UE Configuration ..... 52
C.3.1.2.3.1 Description ..... 52
C.3.1.2.3.2 Resource Definition ..... 52
C.3.1.2.3.3 Resource Standard Methods ..... 52
C.3.1.3 Data Model ..... 54
C.3.1.3.1 General ..... 54
C.3.1.3.2 Structured data types ..... 54
C.3.1.3.2.1 Type: UeConfigDoc ..... 54
C.3.1.3.2.2 Type: UeConfig ..... 55
C.3.1.3.2.3 Type: ValUeIds ..... 55
C.3.1.3.2.4 Type: ImeiRange ..... 55
C.3.1.3.2.5 Type: SnrRange ..... 55
C.3.1.3.3 Simple data types and enumerations ..... 56
C.3.1.3.3.1 Simple data types ..... 56
C.3.1.4 Error Handling ..... 56
C.3.1.5 CDDL Specification ..... 56
C.3.1.5.1 Introduction ..... 56
C.3.1.5.2 CDDL document ..... 56
C.3.1.6 Media Type ..... 57
C.3.1.7 Media Type registration for application/vnd.3gpp.seal-ue-config-info+cbor ..... 57

**Annex D (informative): Change history ..... 59**

History ..... 60

# Foreword

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2 presented to TSG for approval;

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y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.

z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

**shall** indicates a mandatory requirement to do something

**shall not** indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

**should** indicates a recommendation to do something

**should not** indicates a recommendation not to do something

**may** indicates permission to do something

**need not** indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

**can** indicates that something is possible

**cannot** indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

**will** indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**will not** indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

**might** indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

**might not** indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

**is** (or any other verb in the indicative mood) indicates a statement of fact

**is not** (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

# 1 Scope

The present document specifies the protocol aspects for the configuration management capability of SEAL to support vertical applications (e.g. V2X) over the 3GPP system.

The present document is applicable to the User Equipment (UE) supporting the configuration management client functionality as described in 3GPP TS 23.434 [2], to the application server supporting the configuration management server functionality as described in 3GPP TS 23.434 [2] and to the application server supporting the vertical application server (VAL server) functionality as defined in specific vertical application service (VAL service) specification.

NOTE: The specification of the VAL server for a specific VAL service is out of scope for present document.

# 2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TS 23.434: "Service Enabler Architecture Layer for Verticals (SEAL); Functional architecture and information flows".

[3] IETF RFC 4825: "The Extensible Markup Language (XML) Configuration Access Protocol (XCAP)".

[4] OMA OMA-TS-XDM_Core-V2_1-20120403-A: "XML Document Management (XDM) Specification".

[5] 3GPP TS 24.547: "Identity management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification".

[6] IETF RFC 6750: "The OAuth 2.0 Authorization Framework: Bearer Token Usage".

[7] IETF RFC 7159: "The JavaScript Object Notation (JSON) Data Interchange Format".

[8] 3GPP TS 24.229: "IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3".

[9] IETF RFC 5875: "An Extensible Markup Language (XML) Configuration Access Protocol (XCAP) Diff Event Package".

[10] IETF RFC 6050: "A Session Initiation Protocol (SIP) Extension for the Identification of Services".

[11] IETF RFC 6665: "SIP-Specific Event Notification".

[12] IETF RFC 7252: "The Constrained Application Protocol (CoAP)".

[13] IETF RFC 7959: "Block-Wise Transfers in the Constrained Application Protocol (CoAP)".

[14] IETF RFC 7641: "Observing Resources in the Constrained Application Protocol (CoAP)".

[15] IETF RFC 8323: "CoAP (Constrained Application Protocol) over TCP, TLS, and WebSockets".

[16] void

[17] IETF RFC 8949: "Concise Binary Object Representation (CBOR)".

[18] IETF RFC 8610: "Concise Data Definition Language (CDDL): A Notational Convention to Express Concise Binary Object Representation (CBOR) and JSON Data Structures".

[19] Constrained RESTful Environments (CoRE) Parameters at IANA, https://www.iana.org/assignments/core-parameters/core-parameters.xhtml

[20] IETF RFC 9290: "Concise Problem Details for Constrained Application Protocol (CoAP) APIs".

[21] IETF RFC 9177: "Constrained Application Protocol (CoAP) Block-Wise Transfer Options Supporting Robust Transmission".

[22] IETF RFC 3986: "Uniform Resource Identifier (URI): Generic Syntax".

[23] 3GPP TS 29.501: "Principles and Guidelines for Services Definition".

[24] 3GPP TS 23.682: "Architecture Enhancements to facilitate communications with Packet Data Networks and Applications".

[25] IETF RFC 3339: "Date and Time on the Internet: Timestamps".

[26] 3GPP TS 23.003: "Numbering, addressing and identification".

[27] Void

[28] 3GPP TS 24.008: "Mobile Radio Interface Layer 3 specification; Core Network Protocols; Stage 3".

# 3 Definitions of terms and abbreviations

## 3.1 Terms

For the purposes of the present document, the terms given in 3GPP TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in 3GPP TR 21.905 [1].

**SEAL configuration management client**: An entity that provides the client side functionalities corresponding to the SEAL configuration management service.

**SEAL configuration management server**: An entity that provides the server side functionalities corresponding to the SEAL configuration management service.

For the purposes of the present document, the following terms and definitions given in 3GPP TS 23.434 [2] apply:

**SEAL client**
**SEAL server**
**SEAL service**
**VAL server**
**VAL service**
**VAL user**
**Vertical**
**Vertical application**

## 3.2 Abbreviations

For the purposes of the present document, the abbreviations given in 3GPP TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in 3GPP TR 21.905 [1].

MIME Multipurpose Internet Mail Extensions
SCM-C SEAL Configuration Management Client

SCM-S SEAL Configuration Management Server
SEAL Service Enabler Architecture Layer for verticals

# 4 General description

Configuration management is a SEAL service that provides the configuration management related capabilities to one or more vertical applications. The present document enables a SEAL configuration management client (SCM-C) and a VAL server to manage configuration data in a SEAL configuration management server (SCM-S).

# 5 Functional entities

## 5.1 SEAL configuration management client (SCM-C)

The SCM-C functional entity acts as the application client for configuration related transactions.

To be compliant with the HTTP procedures in the present document the SCM-C:

- shall support the role of XCAP client as specified in IETF RFC 4825 [3];
- shall support the role of XDMC as specified in OMA OMA-TS-XDM_Core-V2_1 [4];
- shall support the procedures in clause 6.2.2;
- shall support the procedures in clause 6.2.3;
- shall support the procedures in clause 6.2.4;and
- shall support the procedures in clause 6.2.5.

To be compliant with the CoAP procedures in the present document the SCM-C:

- shall support the role of CoAP client as specified in IETF RFC 7252 [12];
- shall support the capability to observe resources as specified in IETF RFC 7641 [14];
- shall support the block-wise transfer as specified in IETF RFC 7959 [13];
- may support the robust block transfer as specified in IETF RFC 9177 [21];
- should support CoAP over TCP and Websocket as specified in IETF RFC 8323 [15];
- shall support CBOR encoding as specified in IETF RFC 8949 [17];
- shall support the procedures in clause 6.2.2;
- shall support the procedures in clause 6.2.3;
- shall support the procedures in clause 6.2.4;and
- shall support the procedures in clause 6.2.5

NOTE 1: The security mechanism to be supported for the CoAP procedures is described in 3GPP TS 24.547 [5].

NOTE 2: Support for TCP for the CoAP procedures is required if the client connects over the network which blocks or impedes the use of UDP, e.g. when NATs are present in the communication path.

NOTE 3: The CoAP protocol supports mechanism for reliable message exchange over UDP. Use of TCP can also be beneficial if reliable transport is required for other reasons, e.g. better observability of resources. Usage of CoAP over TCP is an implementation choice.

NOTE 4: Support for the robust block transfer mechanism for the CoAP procedures is beneficial in environments where packet loss is highly asymmetrical and where performance optimization of block transfers is required.

## 5.2 SEAL configuration management server (SCM-S)

The SCM-S is a functional entity used to configure one or more vertical applications with 3GPP system related vertical applications provisioning information and configure data on the SEAL configuration management client.

To be compliant with the HTTP procedures in the present document the SCM-S:

- shall support the role of XCAP server as specified in IETF RFC 4825 [3];
- shall support the role of XDMS as specified in OMA OMA-TS-XDM_Core-V2_1 [4];
- shall support the procedures in clause 6.2.2;
- shall support the procedures in clause 6.2.3;
- shall support the procedures in clause 6.2.4;
- shall support the procedures in clause 6.2.5.

To be compliant with the CoAP procedures in the present document the SCM-C:

- shall support the role of CoAP server as specified in IETF RFC 7252 [12];
- shall support the capability to observer resources as specified in IETF RFC 7641 [14];
- shall support the block-wise transfer as specified in IETF RFC 7959 [13];
- shall support the robust block transfer as specified in IETF RFC 9177 [21];
- shall support CoAP over TCP and Websocket as specified in IETF RFC 8323 [15];
- shall support CBOR encoding as specified in IETF RFC 8949 [17];
- shall support the procedures in clause 6.2.2;
- shall support the procedures in clause 6.2.3;
- shall support the procedures in clause 6.2.4;and
- shall support the procedures in clause 6.2.5.

NOTE: The security mechanism to be supported for the CoAP procedures is described in 3GPP TS 24.547 [5]

# 6 Configuration management procedures

## 6.1 General

## 6.2 On-network procedures

### 6.2.1 General

#### 6.2.1.1 Authenticated identity in HTTP request

Upon receiving an HTTP request, the SCM-S shall authenticate the identity of the sender of the HTTP request as specified in 3GPP TS 24.547 [5], and if authentication is successful, the SCM-S shall use the identity of the sender of the HTTP request as an authenticated identity.

#### 6.2.1.2 Authenticated identity in CoAP request

Upon receiving an CoAP request, the SCM-S shall authenticate the identity of the sender of the CoAP request as specified in 3GPP TS 24.547 [5], and if authentication is successful, the SCM-S shall use the identity of the sender of the CoAP request as an authenticated identity.

### 6.2.2 Common procedures

#### 6.2.2.1 Management of configuration update event subscription

##### 6.2.2.1.1 SIP based procedures

###### 6.2.2.1.1.1 General

The VAL service will use the same identity which has been authenticated by VAL service with SIP core using SIP based REGISTER message. If VAL service do not support SIP protocol, then HTTP based method needs to be used.

The SCM-C shall use mechanism provided by VAL service to add access-token in SIP messages. The SCM-S shall identify the originating VAL user ID from the access-token received from SCM-C using the mechanism defined in VAL service specification.

###### 6.2.2.1.1.2 Create subscription

In order to subscribe to notification of changes of one or more group documents of VAL groups identified by VAL group IDs, a SCM-C shall send an initial SIP SUBSCRIBE request to the network according to the UE originating procedures specified in 3GPP TS 24.229 [8] and IETF RFC 5875 [9]. In the initial SIP SUBSCRIBE request, the SCM-C:

a) shall set the Request-URI to the configured public service identity for performing subscription proxy function of the SCM-S;

b) shall include the ICSI value "urn:urn-7:3gpp-service.ims.icsi.seal" (coded as specified in 3GPP TS 24.229 [8]), in a P-Preferred-Service header field according to IETF RFC 6050 [10];

c) shall include the g.3gpp.icsi-ref media feature tag containing the value of "urn:urn-7:3gpp-service.ims.icsi.seal" in the Contact header field;

d) shall include an application/resource-lists+xml MIME body. In the application/resource-lists+xml MIME body, the SCM-C shall include one <entry> element for each configuration document to be subscribed to, such that the "uri" attribute of the <entry> element contains a relative path reference to XCAP URI identifying an XML document to be subscribed to;and

e) if the VAL server wants to fetch the current state only, shall set the Expires header field according to IETF RFC 6665 [11], to zero. Otherwise, shall set the Expires header field to the duration for which VAL user has requested for subscription.

Upon reception of an initial SIP SUBSCRIBE request:

a) with the Event header field set to xcap-diff;

b) with the Request-URI set to own public service identity for performing subscription proxy function of the SCM-S;

c) with an application/resource-lists+xml MIME body; and

d) with the ICSI value "urn:urn-7:3gpp-service.ims.icsi.seal" (coded as specified in 3GPP TS 24 229 [8]), in a P-Asserted-Service header field according to IETF RFC 6050 [10];

the SCM-S:

a) shall identify the originating VAL user ID and shall use the originating VAL user ID as an authenticated identity when performing the authorization;

b) if the authenticated identity is not authorized to subscribe to notification of changes of any resource in the application/resource-lists+xml MIME body, shall reject the request with a SIP 403 (Forbidden) response and shall not continue with rest of the steps;and

c) act as a notifier according to IETF RFC 5875 [9].

#### 6.2.2.1.1.3 Modify subscription

In order to modify or refresh subscription, the SCM-C shall send SIP re-SUBSCRIBE request on the same dialog as the existing subscription, and with the same "Event" header. The SCM-C shall follow the steps specified in clause 6.2.2.1.1.2 to create SIP SUBSCRIBE request.

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with an application/resource-lists+xml MIME body;

the SCM-S:

a) act as a notifier according to IETF RFC 5875 [9].

#### 6.2.2.1.1.4 Delete subscription

In order to delete the subscription, the SCM-C shall send SIP re-SUBSCRIBE request on the same dialog as the existing subscription, and with the same "Event" header. The SCM-C shall follow the steps specified in clause 6.2.2.1.1.2 to create SIP SUBSCRIBE request with following clarification:

a) shall set the Expires header field to zero.

Upon reception of a SIP re-SUBSCRIBE request:

a) with the Event header field set to xcap-diff; and

b) with Expires header field set to zero;

the SCM-S:

a) act as a notifier according to IETF RFC 5875 [9].

#### 6.2.2.1.2 HTTP based procedures

##### 6.2.2.1.2.1 Creating subscription

Upon successful service authorization of the VAL service, the SCM-C shall create a subscription for configuration events by sending an HTTP POST request to the SCM-S. In the HTTP POST request, the SCM-C:

a) shall set the Request URI to the URI of the SCM-S appended with VAL service identity and the value "/configurationEventsSubscription";

b) shall include the Host header with public user identity of SCM-S;

c) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6]; and

c) include the parameters specified in clause A.1.2 serialized into a JavaScript Object Notation (JSON) structure as specified in IETF RFC 7159 [7].

Upon reception of an HTTP POST request from SCM-C where the Request-URI of the HTTP POST request contains "/configurationEventsSubscription", the SCM-S:

a) shall determine the identity of the sender of the received HTTP POST request as specified in clause 6.2.1.1, and:

   1) if the identity of the sender of the received HTTP POST request is not authorized user, shall respond with an HTTP 403 (Forbidden) response to the HTTP POST request and skip rest of the steps;

b) shall generate unique subscription identity and store the subscription details for the authorized user; and

c) shall send an HTTP 200 (OK) response including parameters specified in clause A.1.3.

##### 6.2.2.1.2.2 Modify a subscription

Upon receiving a request from VAL user to modify existing subscription identified with unique subscription identity, the SCM-C:

a) shall generate an HTTP PUT request. In the HTTP PUT request:

   1) shall set the Request URI to the same Request URI used while creating subscription in clause 6.2.2.1.2.1appended with subscription identity;

   2) shall include the Host header with the public user identity of SCM-S;

   3) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6]; and

   4) include the parameters specified in clause A.1.2 serialized into a JavaScript Object Notation (JSON) structure as specified in IETF RFC 7159 [7];and

b) shall send the HTTP PUT request to the SCM-S.

Upon reception of an HTTP PUT request from the SCM-C where the Request-URI of the HTTP PUT request contains "/configurationEventsSubscription" appended with subscription identity, the SCM-S:

a) shall determine the identity of the sender of the received HTTP PUT request as specified in clause 6.2.1.1, and:

   1) if the identity of the sender of the received HTTP PUT request is not authorized user, shall respond with an HTTP 403 (Forbidden) response to the HTTP PUT request and skip rest of the steps;

b) shall determine whether subscription for configuration events exists or not based on received subscription identity in request URI,and:

   1) if subscription does not exist, shall respond with an HTTP 406 (Not Acceptable) response to the HTTP PUT request and skip rest of the steps;

c) shall update the subscription details based on received parameters from the HTTP PUT request; and

d) shall send an HTTP 200 (OK) response including parameters specified in clause A.1.3.

#### 6.2.2.1.2.3 Delete a subscription

Upon receiving a request from VAL user to delete existing subscription identified with unique subscription identity, the SCM-C:

a) shall generate an HTTP DELETE request. In the HTTP DELETE request:

   1) shall set the Request URI to the same Request URI used while creating subscription in clause 6.2.2.1.2.1appended with subscription identity;

   2) shall include the Host header with the public user identity of the SCM-S; and

   3) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6]; and

b) shall send the HTTP DELETE request to the SCM-S.

Upon reception of an HTTP DELETE request from the SCM-C where the Request-URI of the HTTP DELETE request contains "/configurationEventsSubscription" appended with subscription identity, the SCM-S:

a) shall determine the identity of the sender of the received HTTP DELETE request as specified in clause 6.2.1.1, and:

   1) if the identity of the sender of the received HTTP DELETE request is not authorized user, shall respond with an HTTP 403 (Forbidden) response to the HTTP DELETE request and skip rest of the steps;

b) shall determine whether subscription for configuration events exists or not based on received subscription identity in request URI, and

   1) if subscription does not exist, shall respond with an HTTP 406 (Not Acceptable) response to the HTTP DELETE request and skip rest of the steps;

c) shall delete the subscription details based on received parameters from the HTTP DELETE request; and

d) shall send an HTTP 200 (OK) response to the SCM-C.

### 6.2.2.1.3 CoAP based procedures

#### 6.2.2.1.3.1 General

CoAP based procedures shall use the mechanisms to observe a resource as specified in IETF RFC 7641 [14].

NOTE: CoAP "observe" mechanism uses the principle of eventual consistency where an intermediate state change can be lost when UDP is used. If it is critical for the client to receive every change in the resource state (and not just the latest state), TCP can be used to avoid missing notifications.

#### 6.2.2.1.3.2 Create a subscription

In order to subscribe to changes of a configuration document the SCM-C shall send an extended CoAP GET request with the CoAP URI set to the URI of an observable configuration document and with the Observe option set to 0 (Register) as specified in IETF RFC 7641 [14].

Upon reception of such an extended CoAP request from SCM-C where the CoAP URI of the request points at an observable configuration document and with the Observe option set to 0 (Register), the SCM-S:

a) shall perform the steps as for a normal CoAP GET request for a configuration document as defined in clause 6.2.4.4 for VAL UE configuration and in clause 6.2.4.4 for VAL user profile;

b) shall register the SCM-C as an observer as per IETF RFC 7641 [14]; and

c) shall send a CoAP 2.05 (Content) response including the current content of the resource and the Observer option with the initial sequence number of the notifications.

#### 6.2.2.1.3.3 Delete a subscription

In order to unsubscribe from changes of a configuration document the SCM-C shall send a CoAP GET request matching the CoAP GET request used to create the subscription but with the Observe option set to 1 (Deregister) as specified in IETF RFC 7641 [14].

Upon reception of a CoAP GET that matches an active subscription but with the Observe option set to 1 (Deregister), the SCM-S:

a) shall perform the steps as for a normal CoAP GET request for a configuration document as defined in clause 6.2.3.4 for VAL UE configuration and in clause 6.2.4.4 for VAL user profile;

b) shall deregister the SCM-C as an observer as per IETF RFC 7641 [14]; and

c) shall send a CoAP 2.05 (Content) response including the current content of the resource and shall not include the Observer option.

### 6.2.2.2 Notifications

#### 6.2.2.2.1 SIP based procedures

##### 6.2.2.2.1.1 Client procedure

Upon receiving a SIP NOTIFY request associated with a subscription created as result of the sent initial SIP SUBSCRIBE request, the SCM-C:

a) shall handle the SIP NOTIFY request according to IETF RFC 5875 [9].

##### 6.2.2.2.1.2 Server procedure

In order to send notification of group document update event, the SCM-S shall send SIP NOTIFY to SCM-C according to IETF RFC 5875 [9].

#### 6.2.2.2.2 HTTP based procedures

##### 6.2.2.2.2.1 Receiving configuration update notification

Upon receiving an HTTP POST request over a Callback-URI I which was given to SCM-S at time of the configuration update event subscription message, the SCM-C:

a) shall validate the subscription identity received in the "Identity" parameter of the HTTP POST request. If the subscription identity is not valid, the SCM-C:

   1) shall send an HTTP 406 (Not Acceptable) response and skip rest of the steps;

b) shall send an HTTP 200 (OK) message; and

c) shall notify the VAL user about the modification of configuration document based on the "Event" parameter.

Based on VAL user's request, the SCM-C may also retrieve the configuration document as specified in clause 6.2.3 or in clause 6.2.4.

##### 6.2.2.2.2.2 Sending group modify notification

Upon successful modification of VAL user profile document or VAL UE configuration document, the SCM-S sends a notification to SCM-C. The SCM-S:

a) shall check whether valid configuration update event subscription exists for event SUBSCRIBE_USER_PROFILE_MODIFICATION (0x01) OR SUBSCRIBE_UE_CONFIG_MODIFICATION (0x02) as defined in clause A.1.2 or not;

   1) if valid subscription does not exist, shall skip rest of the steps;

b) shall generate an HTTP POST message to notify configuration update notification. In HTTP POST message:

1) shall set the request URI to the Callback-URI received in the creating subscription procedure;

2) shall set the Content-Type header to "application/json"; and

3) shall include an HTTP request entity-body with the parameters specified in clause B.2 serialized into a JavaScript Object Notation (JSON) structure; and

c) shall sent an HTTP POST request towards SCM-C.

#### 6.2.2.2.3 CoAP based procedures

##### 6.2.2.2.3.1 Client procedure

Upon receiving a CoAP 2.05 (Content) response that matches the extended CoAP GET request which initiated the subscription and which contains the Observe option, the SCM-C:

a) shall handle the response according to IETF RFC 7641 [14]; and

b) shall notify the VAL user about the modification of the configuration document.

##### 6.2.2.2.3.2 Server procedure

In order to send a notification when the configuration document is modified, the SCM-S shall send a CoAP 2.05 (Content) response to SCM-C containing the modified document and the Observe option according to IETF RFC 7641 [14]. The Content-Format specified in a 2.xx notification shall be the same as the one used in the initial response to the GET request received for the subscription.

## 6.2.3 VAL UE configuration data

### 6.2.3.1 SCM client HTTP procedure

Upon receiving a request from the VAL user to retrieve a VAL UE configuration data, the SCM-C shall send an HTTP GET request to the SCM-S according to procedures specified in IETF RFC 4825 [3] "*Fetch a Document*". In HTTP GET request, the SCM-C:

a) shall set the Request-URI to a XCAP URI identifying the XML document to be retrieved. In the Request-URI:

1) the "XCAP Root" is set to the URI of the SCM-S;

2) the "auid" is set to specific VAL service identity; and

3) the document selector is set to a document URI pointing to the VAL UE configuration document;

b) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6]; and

c) may include the parameters specified in clause A.2.1 serialized into a JavaScript Object Notation (JSON) structure as specified in IETF RFC 7159 [7]

### 6.2.3.2 SCM server HTTP procedure

Upon reception of an HTTP GET request where the Request-URI of the HTTP GET request identifies a UE configuration document as specified in the specific vertical application, the SCM-S:

a) shall determine the identity of the sender of the received HTTP GET request as specified in clause 6.2.1.1, and:

1) if the identity of the sender of the received HTTP GET request is not authorized to fetch requested configuration document, shall respond with a HTTP 403 (Forbidden) response to the HTTP GET request and skip rest of the steps; and

b) shall support handling an HTTP GET request from a SCM-C according to procedures specified in IETF RFC 4825 [3] "*GET Handling*".

### 6.2.3.3 SCM client CoAP procedure

Upon receiving a request from the VAL user to retrieve a VAL UE configuration data, the SCM-C shall send a CoAP GET request to the SCM-S. In the CoAP GET request, the SCM-C:

a) shall set the CoAP URI identifying the user profile document to be retrieved according to the resource API definition in clause C.3.1:

   1) the "apiRoot" is set to the SCM-S URI;

   2) the "valServiceId" is set to specific VAL service;

   3) if the SCM-C does not know the "ueConfigDocId" of the UE configuration document at the SGM-S, the SCM-C shall make a GET request for the UE Configurations resource as described in clause C.3.1.2.2.3.1 and shall set applicable query parameters defined in table C.3.1.2.2.3.1-1; and

   4) if the SCM-C knows the "ueConfigDocId" of the UE configuration document at the SGM-S, the SCM-C shall make a GET request for the Individual UE Configuration resource as described in clause C.3.1.2.3.3.1, and shall set "ueConfigDocId" to point to the VAL UE configuration document; and

b) shall send the request protected with the relevant ACE profile (OSCORE profile or DTLS profile) as described in 3GPP TS 24.547 [5].

### 6.2.3.4 SCM server CoAP procedure

Upon reception of an CoAP GET request where the CoAP URI of the request identifies the UE Configurations resource as described in clauseC.3.1.2.2.3.1, the SCM-S:

a) shall determine the identity of the sender of the received CoAP GET request as specified in clause 6.2.1.2, and:

   1) if the sender is not authorized to fetch the requested UE configuration document(s), shall respond with a CoAP 4.03 (Forbidden) response to the CoAP GET request and skip rest of the steps;

b) shall support handling a CoAP GET request from a SCM-C according to procedures specified in IETF RFC 7252 [12];

c) shall check if the resource exists for the given VAL service, and:

   1) if the resource does not exist, shall return a 4.04 (Not found) response and skip rest of the steps; and

d) shall return a 2.05 (Content) response including all the UE configuration documents found for the given values of the query parameters defined in table C.3.1.2.2.3.1-1.

Upon reception of an CoAP GET request where the CoAP URI of the request identifies the Individual UE Configuration resource as described in clause C.3.1.2.3.3.1, the SCM-S:

a) shall determine the identity of the sender of the received CoAP GET request as specified in clause 6.2.1.2, and:

   1) if the sender is not authorized to fetch the requested UE configuration document, shall respond with a CoAP 4.03 (Forbidden) response to the CoAP GET request and skip rest of the steps;

b) shall support handling a CoAP GET request from a SCM-C according to procedures specified in IETF RFC 7252 [12]; and

c) shall check if the resource pointed at by the CoAP URI exists and:

   1) if it exists, shall return the UE configuration document in a 2.05 (Content) response; or

2) otherwise, shall return a 4.04 (Not found) response.

## 6.2.4 VAL user profile data

### 6.2.4.1 SCM client HTTP procedure

Upon receiving a request from the VAL user to retrieve a VAL user profile data, the SCM-C shall send an HTTP GET request to the SCM-S according to procedures specified in IETF RFC 4825 [3] "*Fetch a Document*". In HTTP GET request, the SCM-C:

a) shall set the Request-URI to a XCAP URI identifying the XML document to be retrieved. In the Request-URI:

   1) the "XCAP Root" is set to the URI of the SCM-S;

   21) the "auid" is set to specific VAL service identity; and

   3) the document selector is set to a document URI pointing to the VAL user profile document; and

b) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6].

### 6.2.4.2 SCM server HTTP procedure

Upon reception of an HTTP GET request where the Request-URI of the HTTP GET request identifies a user profile document as specified in the specific vertical application, the SCM-S follow the procedure as described in clause 6.2.3.2.

### 6.2.4.3 SCM client CoAP procedure

Upon receiving a request from the VAL user to retrieve a VAL user profile data, the SCM-C shall send a CoAP GET request to the SCM-S. In the CoAP GET request, the SCM-C:

a) shall set the CoAP URI identifying the user profile document to be retrieved according to the resource API definition in clause C.2.1:

   1) the "apiRoot" is set to the SCM-S URI;

   2) the "valServiceId" is set to specific VAL service;

      3) if the SCM-C does not know the "profileDocId" of the user profile document at the SGM-S, the SCM-Cshall use the User Profiles resource GET, as described in clause C.2.1.2.2.3.1, and shall set the val-tgt-ue query parameter to either the VAL user identity or VAL UE identity; and

      4) if the SCM-C knows the "profileDocId" of the user profile document at the SGM-S, the SCM-C shall use the Individual User Profile resource GET, as described in clause C.2.1.2.3.3.1, and shall set "profileDocId" to point to the VAL user profile document; and

b) shall send the request protected with the relevant ACE profile (OSCORE profile or DTLS profile) as described in 3GPP TS 24.547 [5].

### 6.2.4.4 SCM server CoAP procedure

Upon reception of an CoAP GET request where the CoAP URI of the request identifies the User Profiles resource as described in clause C.2.1.2.2.3.1, the SCM-S:

a) shall determine the identity of the sender of the received CoAP GET request as specified in clause 6.2.1.2, and:

   1) if the identity of the sender of the received CoAP GET request is not authorized to fetch requested user profile document(s), shall respond with a CoAP 4.03 (Forbidden) response to the CoAP GET request and skip rest of the steps;

b) shall support handling a CoAP GET request from a SCM-C according to procedures specified in IETF RFC 7252 [12]; and

c) shall check if the resource exists for the given VAL service, and:

1) if the resource does not exist, shall return a 4.04 (Not found) response and skip rest of the steps; and

d) shall return a 2.05 (Content) response including all the user profile documents found for the given VAL user or VAL UE given in the query parameter.

Upon reception of a CoAP GET request where the CoAP URI of the request identifies the Individual User Profile resource as described in clause C.2.1.2.3.3.1, the SCM-S:

a) shall determine the identity of the sender of the received CoAP GET request as specified in clause 6.2.1.2, and:

1) if the identity of the sender of the received CoAP GET request is not authorized to fetch requested user profile document, shall respond with a CoAP 4.03 (Forbidden) response to the CoAP GET request and skip rest of the steps;

b) shall support handling a CoAP GET request from a SCM-C according to procedures specified in IETF RFC 7252 [12]; and

c) shall check if the resource pointed at by the CoAP URI exists and:

1) if it exists, shall return the user profile document in the 2.05 (Content) response; or

2) otherwise, shall return a 4.04 (Not found) response.

## 6.2.5 Update VAL user profile data

### 6.2.5.1 SCM client HTTP procedure

Upon receiving a request from the VAL user to update the VAL user profile configuration document, the SCM-C shall create an XML document as specified in coding of the specific vertical application and shall send the XML document to the SCM-S according to procedures specified in IETF RFC 4825 [3] "*Create or Replace a Document*". In the HTTP POST request, the SCM-C:

a) shall set the Request URI to a XCAP URI identifying an XML document to be updated. In the Request-URI:

1) the "XCAP Root" is set to the URI of the SCM-S;

2) the "auid" is set to specific VAL service identity; and

3) the document selector is set to the VAL user profile;

b) shall include an Authorization header field with the "Bearer" authentication scheme set to an access token of the "bearer" token type as specified in IETF RFC 6750 [6];

c) shall include a Content-Type header field set to "application/vnd.3gpp.seal-user-profile-info+xml"; and

d) shall include an application/vnd.3gpp.seal-user-profile-info+xml MIME body and in the <seal-user-profile> root element:

1) may include <profile-name> element indicating name of the profile;

2) may include <status> element indicating status of the profile;

3) may include <is-default> element indicating that the current profile is the selected profile for the requesting user;

4) shall include <profile-configuration> element as specified in clause 7; and

5) shall include "user-profile-index" attribute indicating the unique profile number.

### 6.2.5.2 SCM server HTTP procedure

Upon reception of an HTTP PUT request where the Request-URI of the HTTP PUT request identifies an XML document as specified in the specific vertical application, the SCM-S:

a) shall determine the identity of the sender of the received HTTP PUT request as specified in clause 6.2.1.1, and:

1) if the identity of the sender of the received HTTP PUT request is not authorized to update the configuration document, shall respond with a HTTP 403 (Forbidden) response to the HTTP PUT request and skip rest of the steps; and

b) shall support receiving an XML document as specified in application usage of the specific vertical application according to procedures specified in IETF RFC 4825 [3] "*PUT Handling*".

#### 6.2.5.3 SCM client CoAP procedure

Upon receiving a request from the VAL user to update the VAL user profile configuration document, the SCM-C shall send a CoAP PUT request to the SCM-S. In the CoAP PUT request, the SCM-C:

a) shall set the CoAP URI identifying the user profile document to be updated according to the resource definition in clause C.2.1.2.3.2:

1) the "apiRoot" is set to the SCM-S URI;

2) the "valServiceId" is set to specific VAL service; and

3) the "profileDocId" to point to the VAL user profile document;

b) shall include Content-Format option set to "application/vnd.3gpp.seal-user-profile-info+cbor";

c) shall include "ProfileDoc" object with "profileInformation" which:

1) may contain "profileName" element indicating name of the profile;

2) shall contain "status" element indicating status of the profile;

3) may contain "isDefault" element indicating that the current profile is the selected profile for the requesting user; and

4) shall contain "profileConfig" elements; and

d) shall send the request protected with the relevant ACE profile (OSCORE profile or DTLS profile) as described in 3GPP TS 24.547 [5].

#### 6.2.5.4 SCM server CoAP procedure

Upon reception of an CoAP PUT request where the CoAP URI of the request identifies Individual User Profile resource as described in clauseC.2.1.2.3.3.2, the SCM-S:

a) shall determine the identity of the sender of the received CoAP PUT request as specified in clause 6.2.1.2, and:

1) if the identity of the sender of the received CoAP PUT request is not authorized to update requested user profile document(s), shall respond with a CoAP 4.03 (Forbidden) response to the CoAP PUT request and skip rest of the steps;

b) shall support handling an CoAP PUT request from a SCM-C according to procedures specified in IETF RFC 7252 [12]; and

c) shall replace the user profile documents pointed at by the CoAP URI with the "ProfileDoc" received in the request.

## 6.3 Off-network procedures

The off-network procedures are out of scope of the present document in this release of the specification.

# 7 Coding

## 7.1 VAL user profile document

### 7.1.1 General

### 7.1.2 Application unique ID

The AUID shall be set to the VAL service ID as specified in specific VAL service specification.

### 7.1.3 Data structure

The <seal-user-profile> element shall be the root element of the VAL user-profile configuration document.

The <seal-user-profile> element:

a) may include a <profile-name> element;

b) may include a <status> element;

c) may include a <is-default> element;

d) shall include a <profile-configuration> element which:

1) may include a <common> element;

2) may include a <on-network> element;

3) may include a <off-network> element; and

e) shall include a "user-profile-index" attribute; and

f) may include any other attribute for the purposes of extensibility.

### 7.1.4 XML Schema

The seal user profile configuration document shall be composed according to the following XML schema:

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  xmlns="urn:3gpp:ns:seal:SealUserProfile:1.0"
  targetNamespace="urn:3gpp:ns:seal:SealUserProfile:1.0"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:sealup="urn:3gpp:ns:seal:SealUserProfile:1.0"
  elementFormDefault="qualified"
  attributeFormDefault="unqualified">

<xs:import namespace="http://www.w3.org/XML/1998/namespace"
  schemaLocation="http://www.w3.org/2001/xml.xsd"/>

  <xs:element name="seal-user-profile">
    <xs:complexType>
      <xs:choice minOccurs="1" maxOccurs="unbounded">
        <xs:element name="profile-name" type="sealup:NameType"/>
        <xs:element name="status" type="xs:boolean"/>
        <xs:element name="is-default" type="xs:boolean"/>
        <xs:element name="profile-configuration" type="sealup:ProfileConfigurationType"/>
        <xs:element name="anyExt" type="sealup:anyExtType" minOccurs="0"/>
        <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
      </xs:choice>
      <xs:attribute name="user-profile-index" type="xs:unsignedByte" use="required"/>
      <xs:anyAttribute namespace="##any" processContents="lax"/>
    </xs:complexType>
  </xs:element>
```

```
  <xs:complexType name="NameType">
    <xs:simpleContent>
      <xs:extension base="xs:token">
        <xs:attribute ref="xml:lang"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>

  <xs:complexType name="ProfileConfigurationType">
    <xs:choice minOccurs="1" maxOccurs="unbounded">
        <xs:element name="common" type="sealup:CommonType"/>
        <xs:element name="on-network" type="sealup:OnNetworkType"/>
        <xs:element name="off-network" type="sealup:OffNetworkType"/>
        <xs:element name="anyExt" type="sealup:anyExtType" minOccurs="0"/>
        <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:choice>
  </xs:complexType>
  <xs:complexType name="CommonType" />
  <xs:complexType name="OnNetworkType" />
  <xs:complexType name="OffNetworkType" />
  <xs:complexType name="anyExtType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

### 7.1.5 Semantics

The <seal-user-profile> element is the root element of the XML document.

The <profile-name> element of <seal-user-profile> element specifies the name of the SEAL user profile configuration document.

The <status> element of <seal-user-profile> element is of type "Boolean" and indicates whether this particular SEAL user profile is enabled or disabled.

The <is-default> element of <seal-user-profile> element is of type "Boolean" and indicates whether this particular SEAL user profile is default profile for VAL user or not.

The "user-profile-index" attribute element of <seal-user-profile> element contains a positive number which provides profile id. This attribute is used only when multiple user-profile for a VAL user is supported.

The <profile-configuration> element of <seal-user-profile> element contains actual profile configuration. The VAL application which uses SEAL user-profile may provide its own profile configuration specific to VAL application.

The VAL service may further extend the <common> element of the <profile-configuration> element of the <seal-user-profile> element to include VAL service specific common user profile configuration.

The VAL service may further extend the <on-network> element of the <profile-configuration> element of the <seal-user-profile> element to include VAL service specific user profile configuration for on-network features.

The VAL service may further extend the <off-network> element of the <profile-configuration> element of the <seal-user-profile> element to include VAL service specific user profile configuration for off-network features.

### 7.1.6 MIME type

The MIME type for VAL user profile configuration shall be set to "vnd.3gpp.seal-user-profile-info+xml".

### 7.1.7 IANA registration template

Your Name:

<MCC name>

Your Email Address:

<MCC email address>

Media Type Name:

Application

Subtype name:

vnd.3gpp.seal-user-profile-info+xml

Required parameters:

None

Optional parameters:

"charset" the parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in section 9.1 of IETF RFC 7303.

Encoding considerations:

binary.

Security considerations:

Same as general security considerations for application/xml media type as specified in section 9.1 of IETF RFC 7303. In addition, this media type provides a format for exchanging information in SIP or in HTTP. So the security considerations from IETF RFC 3261 apply while exchanging information in SIP and the security considerations from IETF RFC 2616 apply while exchanging information in HTTP.

The information transported in this media type does not include active or executable content.

Mechanisms for privacy and integrity protection of protocol parameters exist. Those mechanisms as well as authentication and further security mechanisms are described in 3GPP TS 24.229.

This media type does not include provisions for directives that institute actions on a recipient's files or other resources.

This media type does not include provisions for directives that institute actions that, while not directly harmful to the recipient, may result in disclosure of information that either facilitates a subsequent attack or else violates a recipient's privacy in any way.

This media type does not employ compression.

Interoperability considerations:

Same as general interoperability considerations for application/xml media type as specified in section 9.1 of IETF RFC 7303. Any unknown XML elements and any unknown XML attributes are to be ignored by recipient of the MIME body.

Published specification:

3GPP TS 24.546 "Configuration management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification", available via http://www.3gpp.org/specs/numbering.htm.

Applications Usage:

Applications supporting the SEAL configuration management procedures as described in the published specification.

Fragment identifier considerations:

The handling in section 5 of IETF RFC 7303 applies.

Restrictions on usage:

None

Provisional registration? (standards tree only):

N/A

Additional information:

1. Deprecated alias names for this type: none
2. Magic number(s): none
3. File extension(s): none
4. Macintosh File Type Code(s): none
5. Object Identifier(s) or OID(s): none

Intended usage:

Common

Person to contact for further information:

- Name: <MCC name>
- Email: <MCC email address>
- Author/Change controller:
  i) Author: 3GPP CT1 Working Group/3GPP_TSG_CT_WG1@LIST.ETSI.ORG
  ii) Change controller: <MCC name>/<MCC email address>

## 7.2 VAL UE configuration document

### 7.2.1 General

### 7.2.2 Application unique ID

The AUID shall be set to the VAL service ID as specified in specific VAL service specification.

### 7.2.3 Data structure

The SEAL UE configuration document structure is specified in this clause.

The <seal-UE-configuration> document:

1) shall include a "domain" attribute;
2) shall include a <VAL-UE-id> element;
3) may include a <VAL-service-id> element;
4) may include a <name> element;
5) may include a <common> element;
6) may include an <on-network> element; and
7) may include any other attribute for the purposes of extensibility.

The <VAL-UE-id> element:

1) may contain a list of <instance-ID-URN> elements; and
2) may contain a list of <IMEI-range> elements.

The <IMEI-range> element:

1) shall contain a <TAC> element;

2) may contain a list of <SNR> elements; and

3) may contain <SNR-range> element.

The <SNR-range> element:

1) shall contain a <low-SNR> element; and

2) shall contain a <high-SNR> element.

## 7.2.4 XML schema

```
<?xml version="1.0" encoding="UTF-8"?>
<xs:schema
  xmlns="urn:3gpp:ns:seal:sealUEConfig:1.0"
  targetNamespace="urn:3gpp:ns:seal:sealUEConfig:1.0"
  xmlns:xs="http://www.w3.org/2001/XMLSchema"
  xmlns:sealuec="urn:3gpp:ns:seal:sealUEConfig:1.0"
  elementFormDefault="qualified"
  attributeFormDefault="unqualified">

<xs:import namespace="http://www.w3.org/XML/1998/namespace"
  schemaLocation="http://www.w3.org/2001/xml.xsd"/>


  <xs:element name="seal-UE-configuration">
    <xs:complexType>
      <xs:sequence>
        <xs:choice minOccurs="0" maxOccurs="unbounded">
          <xs:element name="VAL-UE-id" type="sealuec:VALUEIDType"/>
          <xs:element name="VAL-service-id" type="xs:string"/>
          <xs:element name="name" type="sealuec:NameType"/>
        </xs:choice>
        <xs:element name="common" type="sealuec:CommonType"/>
        <xs:element name="on-network" type="sealuec:On-networkType"/>
        <xs:element name="anyExt" type="sealuec:anyExtType" minOccurs="0"/>
        <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
      </xs:sequence>
      <xs:attribute name="domain" type="xs:anyURI" use="required"/>
      <xs:anyAttribute namespace="##any" processContents="lax"/>
    </xs:complexType>
  </xs:element>

  <xs:complexType name="NameType">
    <xs:simpleContent>
      <xs:extension base="xs:token">
        <xs:attribute ref="xml:lang"/>
        <xs:attributeGroup ref="sealuec:IndexType"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>

  <xs:complexType name="VALUEIDType">
    <xs:choice minOccurs="0" maxOccurs="unbounded">
      <xs:element name="instance-ID-URN" type="xs:anyURI"/>
      <xs:element name="IMEI-range" type="sealuec:IMEI-rangeType"/>
      <xs:element name="anyExt" type="sealuec:anyExtType" minOccurs="0"/>
      <xs:any namespace="##other" processContents="lax"/>
    </xs:choice>
    <xs:attributeGroup ref="sealuec:IndexType"/>
    <xs:anyAttribute namespace="##any" processContents="lax"/>
  </xs:complexType>

  <xs:complexType name="IMEI-rangeType">
    <xs:sequence>
      <xs:element name="TAC" type="sealuec:tacType"/>
      <xs:choice minOccurs="0" maxOccurs="unbounded">
        <xs:element name="SNR" type="sealuec:snrType"/>
        <xs:element name="SNR-range" type="sealuec:SNR-rangeType"/>
      </xs:choice>
      <xs:element name="anyExt" type="sealuec:anyExtType" minOccurs="0"/>
      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
```

```
    <xs:attributeGroup ref="sealuec:IndexType"/>
    <xs:anyAttribute namespace="##any" processContents="lax"/>
  </xs:complexType>

  <xs:complexType name="SNR-rangeType">
    <xs:sequence>
      <xs:element name="low-SNR" type="sealuec:snrType"/>
      <xs:element name="high-SNR" type="sealuec:snrType"/>
      <xs:element name="anyExt" type="sealuec:anyExtType" minOccurs="0"/>
      <xs:any namespace="##other" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
    <xs:attributeGroup ref="sealuec:IndexType"/>
    <xs:anyAttribute namespace="##any" processContents="lax"/>
  </xs:complexType>

  <xs:simpleType name="tac-baseType">
      <xs:restriction base="xs:decimal">
        <xs:totalDigits value="8"/>
      </xs:restriction>
  </xs:simpleType>

  <xs:complexType name="tacType">
    <xs:simpleContent>
      <xs:extension base="sealuec:tac-baseType">
        <xs:attributeGroup ref="sealuec:IndexType"/>
        <xs:anyAttribute namespace="##any" processContents="lax"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>

  <xs:simpleType name="snr-baseType">
    <xs:restriction base="xs:decimal">
      <xs:totalDigits value="6"/>
    </xs:restriction>
  </xs:simpleType>

  <xs:complexType name="snrType">
    <xs:simpleContent>
      <xs:extension base="sealuec:snr-baseType">
        <xs:attributeGroup ref="sealuec:IndexType"/>
        <xs:anyAttribute namespace="##any" processContents="lax"/>
      </xs:extension>
    </xs:simpleContent>
  </xs:complexType>

  <xs:complexType name="CommonType" />
  <xs:complexType name="On-networkType" />


  <xs:attributeGroup name="IndexType">
    <xs:attribute name="index" type="xs:token"/>
  </xs:attributeGroup>

  <xs:complexType name="anyExtType">
    <xs:sequence>
      <xs:any namespace="##any" processContents="lax" minOccurs="0" maxOccurs="unbounded"/>
    </xs:sequence>
  </xs:complexType>

</xs:schema>
```

## 7.2.5 Semantics

The "domain" attribute of the <seal-UE-configuration> element contains the domain name of the VAL service.

The <name> element of the <seal-UE- configuration> element contains the user displayable name of the SEAL UE configuration document.

The creator of the SEAL UE configuration document may include an <VAL-UE-id> element in the version of the SEAL UE configuration document that is uploaded to the SCM-S. If an <VAL-UE-id> element is included then the SEAL UE configuration document applies only to the VAL UE(s) identified by the <VAL-UE-id> element. If no <VAL-UE-id> element is included then the SEAL UE configuration document applies to all the VAL UEs of the domain.

The <VAL-service-id> element contains identify of the VAL service for which the configuration document is applicable.

If one or more optional <instance-ID-URN> elements is included in the <VAL-UE-id> element then the SEAL UE configuration document applies to the VAL UE with an instance ID equal to the instance ID contained in the <Instance-ID-URN> element.

The <TAC> element of the <IMEI-range> element contains the Type Allocation Code of the VAL UE.

The optional <SNR> element of the <IMEI-range> element contains the individual serial number uniquely identifying VAL UE within the Type Allocation Code contained in the <TAC> element that the SEAL UE configuration document applies to.

If an optional <SNR-range> element is included within the <IMEI-range> element then the SEAL UE configuration document applies to all VAL UEs within the Type Allocation Code contained in the <TAC> element with the serial number equal or greater than the serial number contained in the <low-SNR> element and less than or equal to the serial number contained in the <high-SNR> element.

If no <SNR> element nor <SNR-range> element is included within the <IMEI-range> element then the SEAL UE configuration document applies to all the VAL UE(s) with the Type Allocation Code contained within the <TAC> element of the <IMEI-range> element.

If no <VAL-UE-id> element is included then the SEAL UE configuration document applies to all VAL UEs of the VAL service identified in the "domain" attribute.

The VAL service may further extend the <common> element of the <seal-UE-configuration> to include VAL service specific common UE configuration.

The VAL service may further extend the <on-network> element of the <seal-UE-configuration> to include VAL service specific UE configuration for on-network features.

### 7.2.6 MIME type

The MIME type for VAL user profile configuration shall be set to "vnd.3gpp.seal-ue-config-info+xml".

### 7.2.7 IANA registration template

Your Name:

<MCC name>

Your Email Address:

<MCC email address>

Media Type Name:

Application

Subtype name:

vnd.3gpp.seal-ue-config-info+xml

Required parameters:

None

Optional parameters:

"charset" the parameter has identical semantics to the charset parameter of the "application/xml" media type as specified in section 9.1 of IETF RFC 7303.

Encoding considerations:

binary.

Security considerations:

Same as general security considerations for application/xml media type as specified in section 9.1 of IETF RFC 7303. In addition, this media type provides a format for exchanging information in SIP or in HTTP. So the security considerations from IETF RFC 3261 apply while exchanging information in SIP and the security considerations from IETF RFC 2616 apply while exchanging information in HTTP.

The information transported in this media type does not include active or executable content.

Mechanisms for privacy and integrity protection of protocol parameters exist. Those mechanisms as well as authentication and further security mechanisms are described in 3GPP TS 24.229.

This media type does not include provisions for directives that institute actions on a recipient's files or other resources.

This media type does not include provisions for directives that institute actions that, while not directly harmful to the recipient, may result in disclosure of information that either facilitates a subsequent attack or else violates a recipient's privacy in any way.

This media type does not employ compression.

Interoperability considerations:

Same as general interoperability considerations for application/xml media type as specified in section 9.1 of IETF RFC 7303. Any unknown XML elements and any unknown XML attributes are to be ignored by recipient of the MIME body.

Published specification:

3GPP TS 24.546 "Configuration management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification", available via http://www.3gpp.org/specs/numbering.htm.

Applications Usage:

Applications supporting the SEAL configuration management procedures as described in the published specification.

Fragment identifier considerations:

The handling in section 5 of IETF RFC 7303 applies.

Restrictions on usage:

None

Provisional registration? (standards tree only):

N/A

Additional information:

1. Deprecated alias names for this type: none
2. Magic number(s): none
3. File extension(s): none
4. Macintosh File Type Code(s): none
5. Object Identifier(s) or OID(s): none

Intended usage:

Common

Person to contact for further information:

- Name: <MCC name>
- Email: <MCC email address>

- Author/Change controller:

  i) Author: 3GPP CT1 Working Group/3GPP_TSG_CT_WG1@LIST.ETSI.ORG

  ii) Change controller: <MCC name>/<MCC email address>

# Annex A (normative): Parameters for different operations

## A.1 Creating configuration update event subscription

### A.1.1 General

The information in this annex provides a normative description of the parameters which will be sent by SCM-C while creating configuration update event subscription and the parameters which will be sent by SCM-S as a response to request for creating subscription.

### A.1.2 Client side parameters

The SCM-C shall convey the following parameters while sending request for creating configuration update event subscription.

**Table A.1.2-1: Client side parameters for creating configuration update event subscription**

| Parameter | Description |
|---|---|
| Callback-URI | REQUIRED. Represents where to send HTTP notifications |
| Subscription Info | REQUIRED. Represents a space-separated list of the subscription type information as specified in table A.1.2-2. |

**Table A.1.2-2: Subscription information**

| Parameter | Description |
|---|---|
| Event | REQUIRED. Represents the type of notification which client requires. This specification defines following type of notifications:<br>- 0x01: SUBSCRIBE_USER_PROFILE_MODIFICATION<br>- 0x02: SUBSCRIBE_UE_CONFIG_MODIFICATION |
| expiry time | REQUIRED. Represents the time in seconds up to which the subscription is desired to be kept active and the time after which the subscribed event shall stop generating notifications. |

### A.1.3 Server side parameters

The SCM-S shall convey the following parameters while sending response to the creating configuration update event subscription request.

**Table A.1.3-1: Server side parameters for response to creating configuration update event subscription**

| Parameter | Description |
|---|---|
| Identity | REQUIRED. A unique string representing subscription identity. |

## A.2 Retrieve VAL UE configuration data

### A.2.1 Client side parameters

The SGM-C shall convey the following parameters, if available, while sending request to retrieve a VAL UE configuration data.

**Table A.2.1-1: Client side parameters to retrieve VAL UE configuration data**

| Parameter | Description |
|---|---|
| VAL UE Information | OPTIONAL. Represents additional UE related information required to identify the configuration data (e.g. device type, device vendor, etc). |

# Annex B (normative): Parameters for notifications

## B.1 General

The information in this annex provides a normative description of the parameters which will be sent by SCM-S while sending different types of notification

## B.2 Configuration update notification

The SCM-S shall convey the following parameters while sending configuration notification to SCM-C.

**Table B.2-1: Parameters for configuration update notification**

| Parameter | Description |
|---|---|
| Identity | REQUIRED. A unique string representing notification channel identity. |
| Event | REQUIRED. Shall be set to one of the event as specified in table A.1.2-2 based on which configuration document is updated. |

# Annex C (normative): CoAP resource representation and encoding

## C.1 General

The information in this annex provides a normative description of CoAP resource representation and encoding.

### C.1.1 Resource URI structure

All API URIs of SEAL-UU APIs shall be specified as follows:

{apiRoot}/<apiName>/<apiVersion>

"apiRoot" is configured by means outside the scope of the present document. It includes one of the schemes ("coaps", "coaps+tcp", "coaps+ws"), host and optional port, and an optional prefix string. "apiName" and "apiVersion" shall be set dependent on the API, as defined in the corresponding clauses below.

All resource URIs specified for SEAL-UU APIs shall be defined relative to the above root API URI.

URIs which differ only in the scheme shall point to the same resource.

NOTE: The "apiVersion" will only be increased if the new API version contains backward incompatible changes.

The root structure may be followed by "apiSpecificSuffixes" that are dependent on the API and are defined separately for each API as resource URI where they apply:

{apiRoot}/<apiName>/<apiVersion>/<apiSpecificSuffixes>

### C.1.2 Use of cache

It is recommended for the SEAL clients and servers to support and use the caching mechanism specified in IETF RFC 7252 [12]. This implies support and use of the Max-Age and ETag options.

### C.1.3 Error handling

Table C.1.3-1 lists response payload types that are applicable to all APIs and as responses for all requests in the present specification unless otherwise specified. The CoAP client shall mandatorily support the processing of the status code for all the applicable methods, when received in a CoAP response message.

**Table C.1.3-1: Response payloads supported for responses to all requests.**

| | Data type | Cardinality | Response Codes (NOTE) | Remarks | Applied Methods |
|---|---|---|---|---|---|
| **Response body** | ProblemDetails | 1 | 4.00 Bad Request | Incorrectparameters were passed in the request. | GET, FETCH, POST PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 4.01 Unauthorized | The client is not authorized. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 4.02 Bad Option | The request could not be understood by the server due to one or more unrecognized or malformed options. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 4.03 Forbidden | This represents the case when the server is able to understand the request but unable to fulfil the request due to errors (e.g. the requested parameters are out of range). More information may be provided in the "invalidParams" attribute of the "ProblemDetails" structure. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 4.04 Not Found | The resource URI was incorrect. | GET, POST, PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 4.06 Not Acceptable | The content format provided in the "Accept" option is not acceptable by the server. | GET, FETCH, |
| | ProblemDetails | 1 | 4.13 Request Entity Too Large | If the received CoAP request contains entity larger than the server is able to process, the server shall reject the CoAP request with this status code. The server should include Size1 option in the response with the maximum size of the request entity it can handle. | FETCH, POST, PUT, PATCH, iPATCH, |
| | ProblemDetails | 1 | 4.15 Unsupported Content-Format | The code indicates that the resource is in a format which is not supported by the server for the method. | FETCH, POST, PUT, PATCH, iPATCH |
| | ProblemDetails | 1 | 4.22 Unprocessable Entity | The code indicates the server is unable to or is incapable of processing the request. | FETCH, PATCH, iPATCH |
| | ProblemDetails | 1 | 4.29 Too Many Requests | The code indicates that due to excessive traffic which, if continued over time, may lead to (or may increase) an overload situation. The CoAP option "Max-Age" may be added in the response to indicate how long the client has to wait before making a new request. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |

| | | | | | |
|---|---|---|---|---|---|
| | ProblemDetails | 1 | 5.00 Internal Server Error | The server encountered an unexpected condition that prevented it from fulfilling the request. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |
| | ProblemDetails | 1 | 5.03 Service Unavailable | The server is unable to handle the request. | GET, FETCH, POST, PUT, PATCH, iPATCH, DELETE |
| NOTE 1: In addition to the above response codes, the CoAP server may also send other valid CoAP response codes, if applicable. The list of all valid CoAP response codes can be found in CoAP Response Code Registry at IANA [19].<br>NOTE 2: CBOR encoding and media type "application/concise-problem-details+cbor" ProblemDetails shall be used for ProblemDetails as defined in the IETF RFC 9290 [20]. | | | | | |

Specific errors are contained in the related API definition for each API.

# C.1.4 Data types applicable to multiple resource representations

## C.1.4.1 General

This clause defines structured data types, simple data types, and enumerations that are applicable to several APIs defined for CoAP resource representations in the present specification and other SEAL specifications and can be referenced from data structures defined in the subsequent clauses and from CoAP resource representations in other SEAL specifications.

NOTE: As a convention, data type names in the present specification follows UpperCamel and parameters follows lowerCamel as specified in clause 5.1.1 of 3GPP TS 29.501 [23].

## C.1.4.2 Referenced structured data types

Table C.1.4.2-1 lists structured data types referenced by multiple CoAP resource representations and defined in this specification or in other specifications.

**Table C.1.4.2-1: Referenced Structured Data Types**

| Data type | Reference | Description |
|---|---|---|
| ValTargetUe | Clause C.2.1.3.2.4 | Information identifying a VAL user ID or VAL UE ID. |
| ScheduledCommunicationTime | Clause C.1.4.4.1 | Defines time schedule for communication. |

## C.1.4.3 Referenced simple data types and enumerations

The simple datatypes based on the CBOR types are defined in table C.1.4.3-1 and the simple data types defined in table C.1.4.3-2 apply to multiple SEAL-UU APIs.

**Table C.1.4.3-1: CBOR-based data types**

| Type name | Description |
|---|---|
| bytes | Is a "byte string" as defined in IETF RFC 8949 [17]. |
| boolean | Is a type which has 2 values "false" and "true" with the values as defined in IETF RFC 8949 [17]. |
| integer | As defined in IETF RFC 8949 [17]. |
| number | Is any number as defined in IETF RFC 8949 [17]. Precision format (half-precision, single-precision, and double-precision) can be indicated. |
| string | Is a "text string" as defined in IETF RFC 8949 [17]. |

**Table C.1.4.3-2: Simple data types applicable to multiple CoAP resource representations**

| Type name | Description |
|---|---|
| Altitude | Number indicating value of altitude in the range from minimum value -32767 to maximum value 32767. |
| Angle | Integer indicating a value of the angle in the range from minimum value 0 to maximum value 360. |
| CellId | String containing a unique identifier of a cell. |
| Confidence | Integer indicating a value of confidence in the range from minimum value 0 to maximum value 100. |
| ExternalGroupId | String containing a local identifier followed by "@" and a domain identifier. Both the local identifier and the domain identifier shall be encoded as strings that do not contain any "@" characters. See clauses 4.6.2 and 4.6.3 of 3GPP TS 23.682 [24] for more information. |
| DateTime | Is a string in the standard format described by the "date-time" production in IETF RFC3339 [25]. |
| DayOfWeek | Integer between and including 1 and 7 denoting a weekday. 1 shall indicate Monday, and the subsequent weekdays shall be indicated with the next higher numbers. 7 shall indicate Sunday. |
| GeographicalAreaId | String identifying a geographical area. |
| InnerRadius | Integer indicating a value of the inner radius in the range from minimum value 0 to maximum value 327675. |
| MbmsSaId | String containing a unique identifier of a MBMS serving area. |
| MbsfnAreaId | String containing a unique identifier of a MBSFN area. |
| Orientation | Integer indicating a value of orientation angle in the range from minimum value 0 to maximum value 180. |
| PlmnId | String containing a unique identifier of a PLMN. |
| TaId | String containing a unique identifier of a tracking area. |
| Tmgi | Byte string containing an identifier of Temporary Mobile Group Identity. The contents of Tmgi are coded as octet 3 and above of Temporary Mobile Group Identity (TMGI) IE as defined in 3GPP TS 24.008 [28] clause 10.5.6.13. |
| TimeOfDay | String with format partial-time or full-time as defined in clause 5.6 of IETF RFC 3339 [25]. Examples, 20:15:00, 20:15:00-08:00 (for 8 hours behind UTC). |
| Uinteger | Unsigned integer, i.e. only value 0 and values above 0 are permissible. |
| Uncertainty | Number indicating value of uncertainty with minimum value 0. |
| Uri | String providing an URI formatted according to IETF RFC 3986 [22]. |

Table C.1.4.3-3 lists simple data types and enumerations referenced by multiple CoAP resource representations defined in this specification or in other specifications.

**Table C.1.4.3-3: Enumerations applicable to multiple CoAP resource representations**

| Type name | Reference | Description |
|---|---|---|
| ConfigType | C.2.1.3.3.1 | Represents the type of configuration. |

## C.1.4.4 Common structured data types

### C.1.4.4.1 Type: ScheduledCommunicationTime

**Table C.1.4.4.1-1: Definition of type ScheduledCommunicationTime**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| daysOfWeek | array(DayOfWeek) | O | 0..6 | Identifies the day(s) of the week. If absent, it indicates every day of the week. | |
| timeOfDayStart | TimeOfDay | O | 0..1 | Identifies the start time of the day. | |
| timeOfDayEnd | TimeOfDay | O | 0..1 | Identifies the end time of the day. | |

### C.1.4.4.2 Type: ProblemDetails

**Table C.1.4.4.2-1: Definition of the ProblemDetails data type**

| Attribute name | Data type | Cardinality | Description |
|---|---|---|---|
| title | string | 0..1 | A short, human-readable summary of the problem type. It should not change from occurrence to occurrence of the problem. |
| detail | string | 0..1 | A human-readable explanation specific to this occurrence of the problem. |
| instance | Uri | 0..1 | A URI reference that identifies the specific occurrence of the problem. |
| cause | string | 0..1 | A machine-readable application error cause specific to this occurrence of the problem.<br>This IE should be present and provide application-related error information, if available. |
| NOTE 1: See IETF RFC 9290 [20] for detailed information and guidance for each attribute.<br>NOTE 2: Additional attributes may be defined per API. | | | |

### C.1.4.4.3 Type: GeographicalCoordinates

**Table C.1.4.4.3-1: Definition of type GeographicalCoordinates**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| lon | number | M | 1 | Longitude expressed as a number with double precision in the range from and including value -180 to and including value 180. |
| lat | number | M | 1 | Latitude expressed as a number with double precision in the range from and including value -90 to and including value 90. |

### C.1.4.4.4 Type: GeographicArea

**Table C.1.4.4.4-1: Definition of type GeographicArea as a list of mutually exclusive alternatives**

| Data type | Cardinality | Discriminator property name | Discriminator mapping | Description |
|---|---|---|---|---|
| Point | 1 | shape | POINT | Geographical area consisting of a single point, represented by its longitude and latitude. |
| PointUncertaintyCircle | 1 | shape | POINT_UNCERTAINTY_CIRCLE | Geographical area consisting of a point and an uncertainty value. |
| PointUncertaintyEllipse | 1 | shape | POINT_UNCERTAINTY_ELLIPSE | Geographical area consisting of a point, plus an uncertainty ellipse and a confidence value. |
| Polygon | 1 | shape | POLYGON | Geographical area consisting of a list of points (between 3 to 15 points). |
| PointAltitude | 1 | shape | POINT_ALTITUDE | Geographical area consisting of a point and an altitude value. |
| PointAltitudeUncertainty | 1 | shape | POINT_ALTITUDE_UNCERTAINTY | Geographical area consisting of a point, an altitude value and an uncertainty value. |
| EllipsoidArc | 1 | shape | ELLIPSOID_ARC | Geographical are consisting of an ellipsoid arc. |

### C.1.4.4.5 Type: Point

**Table C.1.4.4.5-1: Definition of type Point**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POINT". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |

### C.1.4.4.6 Type: PointUncertaintyCircle

**Table C.1.4.4.6-1: Definition of type PointUncertaintyCircle**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POINT_UNCERTAINTY_CIRCLE". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |
| uncertainty | Uncertainty | M | 1 | Indicates the uncertainty value. |

### C.1.4.4.7 Type: PointUncertaintyEllipse

**Table C.1.4.4.7-1: Definition of type PointUncertaintyEllipse**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POINT_UNCERTAINTY_ELLIPSE". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |
| uncertaintyEllipse | UncertaintyEllipse | M | 1 | Indicates an uncertainty ellipse. |
| confidence | Confidence | M | 1 | Indicates the value of confidence. |

### C.1.4.4.8 Type: Polygon

**Table C.1.4.4.8-1: Definition of type Polygon**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POLYGON". |
| pointList | array(GeographicalCoordinates) | M | 3..15 | Array with up to15 items, where each item is a "point". |

### C.1.4.4.9 Type: PointAltitude

**Table C.1.4.4.9-1: Definition of type PointAltitude**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POINT_ALTITUDE". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |
| altitude | Altitude | M | 1 | Indicates the value of altitude. |

### C.1.4.4.10 Type: PointAltitudeUncertainty

**Table C.1.4.4.10-1: Definition of type PointAltitudeUncertainty**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "POINT_ALTITUDE_UNCERTAINTY". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |
| altitude | Altitude | M | 1 | Indicates the value of altitude. |
| uncertaintyEllipse | UncertaintyEllipse | M | 1 | Indicates the uncertainty ellipse |
| uncertaintyAltitude | Uncertainty | M | 1 | Indicates the uncertainty of the altitude. |
| confidence | Confidence | M | 1 | Indicates the value of confidence. |

### C.1.4.4.11 Type: EllipsoidArc

**Table C.1.4.4.11-1: Definition of type EllipsoidArc**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| shape | SupportedGADShapes | M | 1 | It shall take the value "ELLIPSOID_ARC". |
| point | GeographicalCoordinates | M | 1 | Indicates a geographic point represented by its longitude and latitude. |
| innerRadius | InnerRadius | M | 1 | Indicates the value of inner radius of the Ellipsoid Arc. |
| uncertaintyRadius | Uncertainty | M | 1 | Indicates the uncertainty radius of the Ellipsoid Arc. |
| offsetAngle | Angle | M | 1 | Indicates the offset angle of the Ellipsoid Arc. |
| includedAngle | Angle | M | 1 | Indicates the included angle of the Ellipsoid Arc. |
| confidence | Confidence | M | 1 | Indicates the value of confidence. |

#### C.1.4.4.12 Type: UncertaintyEllipse

**Table C.1.4.4.12-1: Definition of type UncertaintyEllipse**

| Attribute name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| semiMajor | Uncertainty | M | 1 | Indicates the semi-major axis of the uncertainty ellipse. |
| semiMinor | Uncertainty | M | 1 | Indicates the semi-minor axis of the uncertainty ellipse. |
| orientationMajor | Orientation | M | 1 | Indicates the orientation angle of the major axis. |

### C.1.4.5 Common enumerations

#### C.1.4.5.1 Enumeration: SupportedGADShapes

**Table C.1.4.5.1-1: Enumeration SupportedGADShapes**

| Enumeration value | Description |
|---|---|
| POINT | Ellipsoid Point |
| POINT_UNCERTAINTY_CIRCLE | Ellipsoid point with uncertainty circle |
| POINT_UNCERTAINTY_ELLIPSE | Ellipsoid point with uncertainty ellipse |
| POLYGON | Polygon |
| POINT_ALTITUDE | Ellipsoid point with altitude |
| POINT_ALTITUDE_UNCERTAINTY | Ellipsoid point with altitude and uncertainty ellipsoid |
| ELLIPSOID_ARC | Ellipsoid Arc |
| LOCAL_2D_POINT_UNCERTAINTY_ELLIPSE | Local 2D point with uncertainty ellipse |
| LOCAL_3D_POINT_UNCERTAINTY_ELLIPSOID | Local 3D point with uncertainty ellipsoid |

## C.2 Resource representation and APIs for VAL user profile

### C.2.1 SU_UserProfile API

#### C.2.1.1 API URI

The CoAP URIs used in CoAP requests from SCM-C towards the SCM-S shall have the Resource URI structure as defined in clause C.1.1 with the following clarifications:

- the <apiName> shall be "su-up";
- the <apiVersion> shall be "v1"; and
- the <apiSpecificSuffixes> shall be set as described in clause C.2.1.2.

## C.2.1.2 Resources

### C.2.1.2.1 Overview

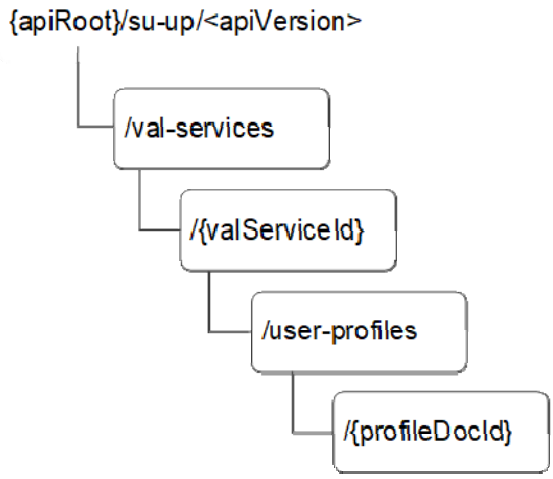



**Figure C.2.1.2.1-1: Resource URI structure of the SU_UserProfile API**

Table C.2.1.2.1-1 provides an overview of the resources and applicable CoAP methods.

**Table C.2.1.2.1-1: Resources and methods overview**

| Resource name | Resource URI | CoAP method | Description |
|---|---|---|---|
| User Profiles | /val-services/{valServiceId}/user-profiles | GET | Retrieve VAL user or VAL UE's user profiles for a given VAL service, according to query criteria. |
| | | POST | Create user profile. |
| Individual User Profile | /val-services/{valServiceId}/user-profiles/{profileDocId} | GET | Retrieve an individual user profile. |
| | | PUT | Update an individual user profile. |
| | | DELETE | Delete an individual user profile. |

### C.2.1.2.2 Resource: User Profiles

#### C.2.1.2.2.1 Description

The User Profiles resource allows a SCM-C to retrieve all the user profiles of a VAL user or a VAL UE for a specific VAL service that are available at a given SCM-S, or allows to create a new user profile.

#### C.2.1.2.2.2 Resource Definition

Resource URI: **{apiRoot}/su-up/<apiVersion>/val-services/{valServiceId}/user-profiles**

This resource shall support the resource URI variables defined in the table C.2.1.2.2.2-1.

**Table C.2.1.2.2.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | string | See clause C.1.1 |
| apiVersion | string | See clause C.2.1.1 |
| valServiceId | string | Identifier of a VAL service. |

#### C.2.1.2.2.3 Resource Standard Methods

##### C.2.1.2.2.3.1 GET

This operation retrieves VAL **u**ser or VAL UE profile information satisfying the filter criteria.

This method shall support the URI query parameters specified in table C.2.1.2.2.3.1-1.

**Table C.2.1.2.2.3.1-1: URI query parameters supported by the GET Request on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| val-tgt-ue | ValTargetUe | M | 1 | Identifies a VAL target UE. |

This method shall support the response data structures and response codes specified in table C.2.1.2.2.3.1-2.

**Table C.2.1.2.2.3.1-2: Data structures supported by the GET Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| array(ProfileDoc) | M | 0..N | 2.05 Content | List of VAL user / VAL UE profile documents. This response shall include user profile information matching the query parameters provided in the request. |
| NOTE: The mandatory CoAP error status codes for the GET Request listed in table C.1.3-1 shall also apply. | | | | |

##### C.2.1.2.2.3.2 POST

This operation creates a VAL user or VAL UE profile information at the SCM-S for a given VAL service.

This method shall support the request data structures specified in table C.2.1.2.2.3.2-1, the response data structures and response codes specified in table C.2.1.2.2.3.2-2, and the response options specified in table C.2.1.2.2.3.2-3.

**Table C.2.1.2.2.3.2-1: Data structures supported by the POST Request payload on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ProfileDoc | M | 1 | The user profile document to be created for a VAL user or VAL UE. |

**Table C.2.1.2.2.3.2-2: Data structures supported by the POST Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ProfileDoc | O | 0..1 | 2.01 Created | The user profile was created successfully.<br><br>The "profileDocId" of the created resource shall be returned in the "Location-Path" option. |
| NOTE: The mandatory CoAP error status codes for the POST method listed in table C.1.3-1 shall also apply. | | | | |

**Table C.2.1.2.2.3.2-3: Options supported by the 2.01 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location-Path | string | M | 1 | Contains the location path of the newly created resource relative to the request URI.<br>It contains the profileDocId segment of the complete resource URI according to the structure: {apiRoot}/su-up/<apiVersion>/val-services/{valServiceId}/user-profiles/{profileDocId} |

### C.2.1.2.3 Resource: Individual User Profile

#### C.2.1.2.3.1 Description

The Individual User Profile resource represents an individual user profile that is created at the SCM-S for a given VAL service. This resource is observable.

#### C.2.1.2.3.2 Resource Definition

Resource URI: **{apiRoot}/su-up/<apiVersion>/val-services/{valServiceId}/user-profiles/{profileDocId}**

This resource shall support the resource URI variables defined in the table C.2.1.2.3.2-1.

**Table C.2.1.2.3.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | string | See clause C.1.1 |
| apiVersion | string | See clause C.2.1.1 |
| valServiceId | string | Identifier of a VAL service. |
| profileDocId | string | Represents an individual user profile resource. |

#### C.2.1.2.3.3 Resource Standard Methods

##### C.2.1.2.3.3.1 GET

This operation retrieves the user profile document.

This method shall support the request options specified in table C.2.1.2.3.3.1-1, the response data structures and response codes specified in table C.2.1.2.3.3.1-2, and the response options specified in table C.2.1.2.3.3.1-3.

**Table C.2.1.2.3.3.1-1: Options supported by the GET Request on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Observe | Uinteger | O | 0..1 | When set to 0 (Register) it extends the GET request to subscribe to the changes of this resource.<br>When set to 1 (Deregister) it cancels the subscription. |
| NOTE: Other request options also apply in accordance with normal CoAP procedures. | | | | |

**Table C.2.1.2.3.3.1-2: Data structures supported by the GET Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ProfileDoc | M | 1 | 2.05 Content | The User profile information based on the request from the SCM-C. |
| NOTE: The mandatory CoAP error status codes for the GET Request listed in table C.1.3-1 shall also apply. | | | | |

**Table C.2.1.2.3.3.1-3: Options supported by the 2.05 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Observe | Uinteger | O | 0..1 | Sequence number of the notification. |
| NOTE: Other response options also apply in accordance with normal CoAP procedures. | | | | |

##### C.2.1.2.3.3.2 PUT

This operation updates the user profile document.

This method shall support the request data structures specified in table C.2.1.2.3.3.2-1 and the response data structures and response codes specified in table C.2.1.2.3.3.2-2.

**Table C.2.1.2.3.3.2-1: Data structures supported by the PUT Request payload on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| ProfileDoc | M | 1 | Updated details of the user profile document. |

**Table C.2.1.2.3.3.2-2: Data structures supported by the PUT Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| ProfileDoc | O | 0..1 | 2.04 Changed | The user profile document updated successfully and the updated user profile document may be returned in the response. |
| NOTE: The mandatory CoAP error status codes for the PUT method listed in table C.1.3-1 shall also apply. | | | | |

C.2.1.2.3.3.3 DELETE

This operation deletes the user profile document.

This method shall support the response data structures and response codes specified in table C.2.1.2.3.3.3-1.

**Table C.2.1.2.3.3.3-1: Data structures supported by the DELETE Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 2.02 Deleted | The individual User profile document matching the profileDocId is deleted. |
| NOTE: The mandatory CoAP error status codes for the DELETE method listed in table C.1.3-1 shall also apply. | | | | |

## C.2.1.3 Data Model

### C.2.1.3.1 General

Table C.2.1.3.1-1 specifies the data types defined specifically for the SU_UserProfile API service.

**Table C.2.1.3.1-1: SU_UserProfile API specific Data Types**

| Data type | Section defined | Description | Applicability |
|---|---|---|---|
| ProfileDoc | C.2.1.3.2.1 | Profile information associated with VAL user ID or VAL UE ID. | |
| ProfileInfo | C.2.1.3.2.2 | Profile information including profile configurations. | |
| ProfileConfig | C.2.1.3.2.3 | Profile configuration including configuration data. | |
| ConfigType | C.2.1.3.3.1 | Specifies type of features for which the configuration data is applicable. | |
| ValTargetUe | C.2.1.3.2.4 | Information identifying a VAL user ID or VAL UE ID. | |

### C.2.1.3.2 Structured data types

#### C.2.1.3.2.1 Type: ProfileDoc

**Table C.2.1.3.2.1-1: Definition of type ProfileDoc**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| profileDocId | string | O | 0..1 | Contains the profileDocId of the complete resource URI of this user profile document according to the structure: {apiRoot}/su-up/<apiVersion>/val-services/{valServiceId}/user-profiles/{profileDocId}<br>This attribute shall be provided by the SCM-S in CoAP responses. | |
| profileInformation | ProfileInfo | M | 1 | Profile information associated with a VAL user or a VAL UE as specified in valTgtUe. | |
| valTgtUe | ValTargetUe | M | 1 | Unique identifier of a VAL user or a VAL UE. | |

#### C.2.1.3.2.2 Type: ProfileInfo

**Table C2.1.4.2.2-1: Definition of type ProfileInfo**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| profileName | string | O | 0..1 | Name of the user profile. | |
| status | boolean | M | 1 | Indicates whether the user profile is enabled or disabled. | |
| profileConfigs) | Array(ProfileConfig) | O | 1..N | List of profile configurations. | |
| isDefault | boolean | O | 0..1 | Indicates whether the user profile is the default profile for VAL user or not. | |

#### C.2.1.3.2.3 Type: ProfileConfig

**Table C.2.1.3.2.3-1: Definition of type ProfileConfig**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| configType | ConfigType | M | 1 | Indicates the type of the profile configuration. | |
| configData | string | M | 1 | Actual user profile configuration data. | |

#### C.2.1.3.2.4 Type: ValTargetUe

**Table C.2.1.3.2.4-1: Definition of type ValTargetUe**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| valUserId | string | O | 0..1 | Unique identifier of a VAL user. | |
| valUeId | string | O | 0..1 | Unique identifier of a VAL UE. | |
| NOTE: Either "valUserId" or "valUeId" shall be present. | | | | | |

#### C.2.1.3.3 Simple data types and enumerations

##### C.2.1.3.3.1 Enumeration: ConfigType

**Table C.2.1.3.3.1-1: Enumeration ConfigType**

| Enumeration value | Description | Applicability |
|---|---|---|
| COMMON | Indicates VAL service specific configuration for common features. | |
| ON_NETWORK | Indicates VAL service specific configuration for on-network features. | |
| OFF_NETWORK | Indicates VAL service specific configuration for off-network features. | |

### C.2.1.4 Error Handling

General error responses are defined in clause C.1.3.

### C.2.1.5 CDDL Specification

#### C.2.1.5.1 Introduction

The data model described in clause C.2.1.3 shall be binary encoded in the CBOR format as described in IETF RFC 8949 [17].

Clause C.2.1.5.2 uses the Concise Data Definition Language described in IETF RFC 8610 [18] and provides corresponding representation of the SU_UserProfile API data model.

#### C.2.1.5.2 CDDL document

```
;;; ProfileDoc
;;+ Represents user profile information associated with a VAL user ID or a VAL UE ID.

ProfileDoc = {
 ? profileDocId: tstr
 profileInformation: ProfileInfo
 valTgtUe: ValTargetUe
 * tstr => any
}

;;; ValTargetUe
;;+ Represents information identifying a VAL user ID or a VAL UE ID.

ValTargetUe = {
 (
 valUserId: tstr                ; Unique identifier of a VAL user.
 //
 valUeId: tstr                  ; Unique identifier of a VAL UE.
 )
}

;;; ProfileInfo
;;+ User profile information.

ProfileInfo = {
 ? profileName: tstr            ; Name of the profile
 status: bool                   ; Indicates whether the user profile is enabled or disabled.
 ? profileConfigs: [+ ProfileConfig]
 ? isDefault: bool              ; Indicates whether the user profile is the default profile for VAL
user or not.
 * tstr => any
}

;;; ProfileConfig
;;+ Profile configuration.

ProfileConfig = {
```

```
 configType: ConfigType
 configData: tstr                ; Actual user profile configuration data.
 * tstr => any
}

;;; ConfigType
;;+ Indicates the type of the configuration.

ConfigType = "COMMON" / "ON_NETWORK" / "OFF_NETWORK" / tstr ; tstr value provides forward-
compatibility with future extensions to the enumeration but is not used to encode content defined in
the present version of this API.
```

### C.2.1.6 Media Type

The media type for a user profile document shall be "application/vnd.3gpp.seal-user-profile-info+cbor".

### C.2.1.7 Media Type registration for application/vnd.3gpp.seal-user-profile-info+cbor

Type name: application

Subtype name: vnd.3gpp.seal-user-profile-info+cbor

Required parameters: none

Optional parameters: none

Encoding considerations: Must be encoded as using IETF RFC 8949 [17]. See 3GPP TS 24.546 clause C.2.1.3 for details.

Security considerations: See Section 10 of IETF RFC 8949 [17] and Section 11 of IETF RFC 7252 [12].

Interoperability considerations: Applications must ignore any key-value pairs that they do not understand. This allows backwards-compatible extensions to this specification.

Published specification: 3GPP TS 24.546 "Configuration management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification", available via http://www.3gpp.org/specs/numbering.htm.

Applications that use this media type: Applications supporting the SEAL configuration management procedures as described in the published specification.

Fragment identifier considerations: Fragment identification is the same as specified for "application/cbor" media type in IETF RFC 8949 [17]. Note that currently that RFC does not define fragmentation identification syntax for "application/cbor".

Additional information:

Deprecated alias names for this type: N/A

Magic number(s): N/A

File extension(s): none

Macintosh file type code(s): none

Person & email address to contact for further information: <MCC name>, <MCC email address>

Intended usage: COMMON

Restrictions on usage: None

Author: 3GPP CT1 Working Group/3GPP_TSG_CT_WG1@LIST.ETSI.ORG

Change controller: <MCC name>/<MCC email address>

# C.3 Resource representation and APIs for UE configuration

## C.3.1 SU_UeConfig API

### C.3.1.1 API URI

The CoAP URIs used in CoAP requests from SCM-C towards the SCM-S shall have the Resource URI structure as defined in clause C.1.1 with the following clarifications:

- the <apiName> shall be "su-uc";
- the <apiVersion> shall be "v1"; and
- the <apiSpecificSuffixes> shall be set as described in clause C.3.1.2.

### C.3.1.2 Resources

#### C.3.1.2.1 Overview

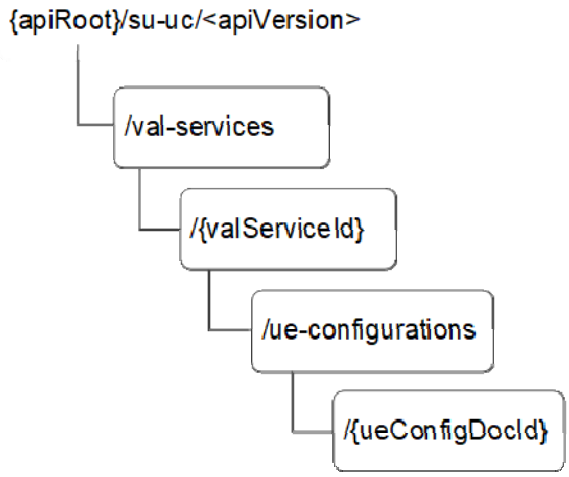



**Figure C.3.1.2.1-1: Resource URI structure of the SU_UeConfig API**

Table C.3.1.2.1-1 provides an overview of the resources and applicable CoAP methods.

**Table C.3.1.2.1-1: Resources and methods overview**

| Resource name | Resource URI | CoAP method | Description |
|---|---|---|---|
| UE Configurations | /val-services/{valServiceId}/ue-configurations | GET | Retrieve UE configurations for a given VAL service, according to query criteria. |
| | | POST | Create UE configuration. |
| Individual UE Configuration | /val-services/{valServiceId}/ue-configurations/{ueConfigDocId} | GET | Retrieve an individual UE configuration. |
| | | PUT | Update an individual UE configuration. |
| | | DELETE | Delete an individual UE configuration. |

Editor's note: Whether any changes required in the API along with its data model based on limitations of constrained devices is FFS.

## C.3.1.2.2 Resource: UE Configurations

### C.3.1.2.2.1 Description

The UE Configurations resource allows a SCM-C to retrieve all the UE configurations of a VAL service domain (e.g. based on device type, device vendor, device number, etc) for a specific VAL service that are available at a given SCM-S.

### C.3.1.2.2.2 Resource Definition

Resource URI: **{apiRoot}/su-uc/<apiVersion>/val-services/{valServiceId}/ue-configurations**

This resource shall support the resource URI variables defined in the table C.3.1.2.2.2-1.

**Table C.3.1.2.2.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | string | See clause C.1.1 |
| apiVersion | string | See clause C3.1.1 |
| valServiceId | string | Identifier of a VAL service. |

### C.3.1.2.2.3 Resource Standard Methods

#### C.3.1.2.2.3.1 GET

This operation retrieves UE configurations satisfying the query criteria.

This method shall support the URI query parameters specified in table C.3.1.2.2.3.1-1.

**Table C.3.1.2.2.3.1-1: URI query parameters supported by the GET Request on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| ue-vendor | string | O | 0..1 | Identity of the UE vendor. |
| ue-type | TypeAllocationCode | O | 0..1 | Type of the UE. |
| ue-snr | SerialNumber | O | 0..1 | Serial number of the UE. |
| ue-uri | Uri | O | 0..1 | URI of the UE. |

This method shall support the response data structures and response codes specified in table C.3.1.2.2.3.1-2.

**Table C.3.1.2.2.3.1-2: Data structures supported by the GET Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| array(UeConfigDoc) | M | 0..N | 2.05 Content | List of UE configuration documents matching any of the query parameters provided in the request. If no query parameters are given, all the UE configuration documents are returned. |
| NOTE: The mandatory CoAP error status codes for the GET Request listed in table C.1.3-1 shall also apply. | | | | |

#### C.3.1.2.2.3.2 POST

This operation creates a UE configuration at the SCM-S for a given VAL service.

This method shall support the request data structures specified in table C.3.1.2.2.3.2-1, the response data structures and response codes specified in table C.3.1.2.2.3.2-2, and the response options specified in table C.3.1.2.2.3.2-3.

**Table C.3.1.2.2.3.2-1: Data structures supported by the POST Request payload on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeConfigDoc | M | 1 | The UE configuration to be created. |

**Table C.3.1.2.2.3.2-2: Data structures supported by the POST Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeConfigDoc | O | 0..1 | 2.01 Created | The UE configuration was created successfully.<br><br>The "ueConfigDocId" of the created resource shall be returned in the "Location-Path" option. |
| NOTE: The mandatory CoAP error status codes for the POST method listed in table C.1.3-1 shall also apply. | | | | |

**Table C.3.1.2.2.3.2-3: Options supported by the 2.01 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Location-Path | string | M | 1 | Contains the location path of the newly created resource relative to the request URI.<br>It contains the ueConfigDocId segment of the complete resource URI according to the structure: {apiRoot}/su-uc/<apiVersion>/val-services/{valServiceId}/ue-configurations/{ueConfigDocId} |

## C.3.1.2.3 Resource: Individual UE Configuration

### C.3.1.2.3.1 Description

The Individual UE Configuration resource represents an individual UE configuration stored at the SCM-S for a given VAL service. This resource is observable.

### C.3.1.2.3.2 Resource Definition

Resource URI: **{apiRoot}/su-uc/<apiVersion>/val-services/{valServiceId}/ue-configurations/{ueConfigDocId}**

This resource shall support the resource URI variables defined in the table C.3.1.2.3.2-1.

**Table C.3.1.2.3.2-1: Resource URI variables for this resource**

| Name | Data Type | Definition |
|---|---|---|
| apiRoot | string | See clause C.1.1 |
| apiVersion | string | See clause C.2.1.1 |
| valServiceId | string | Identifier of a VAL service. |
| ueConfigDocId | string | Represents an individual UE configuration resource. |

### C.3.1.2.3.3 Resource Standard Methods

#### C.3.1.2.3.3.1 GET

This operation retrieves the UE configuration document.

This method shall support the request options specified in table C.3.1.2.3.3.1-1, the response data structures and response codes specified in table C.3.1.2.3.3.1-2, and the response options specified in table C.3.1.2.3.3.1-3.

**Table C.3.1.2.3.3.1-1: Options supported by the GET Request on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Observe | Uinteger | O | 0..1 | When set to 0 (Register) it extends the GET request to subscribe to the changes of this resource.<br>When set to 1 (Deregister) it cancels the subscription. |
| NOTE: Other request options also apply in accordance with normal CoAP procedures. | | | | |

**Table C.3.1.2.3.3.1-2: Data structures supported by the GET Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeConfigDoc | M | 1 | 2.05 Content | The UE configuration based on the request from the SCM-C. |
| NOTE: The mandatory CoAP error status codes for the GET Request listed in table C.1.3-1 shall also apply. | | | | |

**Table C.3.1.2.3.3.1-3: Options supported by the 2.05 Response Code on this resource**

| Name | Data type | P | Cardinality | Description |
|---|---|---|---|---|
| Observe | Uinteger | O | 0..1 | Sequence number of the notification. |
| NOTE: Other response options also apply in accordance with normal CoAP procedures. | | | | |

### C.3.1.2.3.3.2 PUT

This operation updates the UE configuration document.

This method shall support the request data structures specified in table C.3.1.2.3.3.2-1 and the response data structures and response codes specified in table C.3.1.2.3.3.2-2.

**Table C.3.1.2.3.3.2-1: Data structures supported by the PUT Request payload on this resource**

| Data type | P | Cardinality | Description |
|---|---|---|---|
| UeConfigDoc | M | 1 | Updated details of the UE configuration document. |

**Table C.3.1.2.3.3.2-2: Data structures supported by the PUT Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| UeConfigDoc | O | 1 | 2.04 Changed | The UE configuration document updated successfully and the updated UE configuration document may be returned in the response. |
| NOTE: The mandatory CoAP error status codes for the PUT method listed in table C.1.3-1 shall also apply. | | | | |

### C.3.1.2.3.3.3 DELETE

This operation deletes the UE configuration document.

This method shall support the response data structures and response codes specified in table C.3.1.2.3.3.3-1.

**Table C.3.1.2.3.3.3-1: Data structures supported by the DELETE Response payload on this resource**

| Data type | P | Cardinality | Response codes | Description |
|---|---|---|---|---|
| n/a | | | 2.02 Deleted | The individual UE configuration document matching the ueConfigDocId is deleted. |
| NOTE: The mandatory CoAP error status codes for the DELETE method listed in table C.1.3-1 shall also apply. | | | | |

## C.3.1.3 Data Model

### C.3.1.3.1 General

Table C.3.1.3.1-1 specifies the data types defined specifically for the SU_UeConfig resource representation.

**Table C.3.1.3.1-1: SU_UeConfig API specific data types**

| Data type | Section defined | Description | Applicability |
|---|---|---|---|
| UeConfigDoc | C.3.1.3.2.1 | UE configuration document. | |
| UeConfig | C.3.1.3.2.2 | UE configuration including configuration data. | |
| ValUeIds | C.3.1.3.2.3 | VAL UE identifiers. | |
| ImeiRange | C.3.1.3.2.4 | Range of IMEIs. | |
| SnrRange | C.3.1.3.2.5 | Range of UE serial numbers. | |
| SerialNumber | C.3.1.3.3.1 | Serial number of a UE. | |
| TypeAllocationCode | C.3.1.3.3.1 | Type allocation code. | |

Table C.3.1.3.1-2 specifies data types re-used by the SU_UeConfig API service:

**Table C.3.1.3.1-2: Reused data types**

| Data type | Reference | Comments | Applicability |
|---|---|---|---|
| ConfigType | C.2.1.3.3.1 | Configuration type. | |
| Uri | C.1.4.3 | Unified resource identifier. | |

### C.3.1.3.2 Structured data types

#### C.3.1.3.2.1 Type: UeConfigDoc

**Table C.3.1.3.2.1-1: Definition of type UeConfigDoc**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| ueConfigDocId | string | O | 0..1 | Contains the ueConfigDocId of the complete resource URI of this UE configuration document according to the structure: {apiRoot}/su-uc/<apiVersion>/val-services/{valServiceId}/ue-configurations/{ueConfigDocId} This attribute shall be provided by the SCM-S in CoAP responses. | |
| configName | string | O | | Displayable name of the UE configuration document. | |
| valServiceDomain | string | M | 1 | Domain name of the VAL service for which the configuration document is applicable. | |
| valServiceId | string | O | 0..1 | VAL service identity for which the configuration document is applicable. | |
| valUeIds | ValUeIds | O | 0..1 | Defines a set of VAL UE IDs for which the configuration document is applicable. | |
| ueConfigs | array(UeConfig) | O | 1..N | List of UE configurations of different configuration types, i.e. there shall not be 2 configuration with the same value of configType. | |

#### C.3.1.3.2.2 Type: UeConfig

**Table C.3.1.3.2.2-1: Definition of type UeConfig**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| configType | ConfigType (NOTE) | M | 1 | Indicates the type of the UE configuration. | |
| configData | string | M | 1 | Actual UE configuration data. | |
| NOTE: Only the values COMMON and ON_NETWORK are applicable in the present specification. | | | | | |

#### C.3.1.3.2.3 Type: ValUeIds

**Table C.3.1.3.2.3-1: Definition of type ValUeIds**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| uris | array(Uri) | O | 1..N | List of VAL UE identities, each identity defined by a URI. | |
| imeiRanges | array(ImeiRange) | O | 1..N | List of IMEI ranges. | |

#### C.3.1.3.2.4 Type: ImeiRange

**Table C.3.1.3.2.4-1: Definition of type ImeiRange**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| tac | TypeAllocationCode | M | 1 | Type allocation code of the UEs. | |
| snrs | array(SerialNumber) | O | 1..N | List of UE serial numbers. | |
| snrRange | SnrRange | O | 0..1 | Range of UE serial numbers. | |

#### C.3.1.3.2.5 Type: SnrRange

**Table C.3.1.3.2.5-1: Definition of type SnrRange**

| Attribute name | Data type | P | Cardinality | Description | Applicability |
|---|---|---|---|---|---|
| low | SerialNumber | M | 1 | First UE serial number identifying the start of a UE serial number range. | |
| high | SerialNumber | M | 1 | Last UE serial number identifying the end of a UE serial number range. | |

#### C.3.1.3.3 Simple data types and enumerations

##### C.3.1.3.3.1 Simple data types

**Table C.3.1.3.3.1-1: Simple data types**

| Type Name | Type Definition | Description |
|---|---|---|
| TypeAllocationCode | string | Type Allocation Code (TAC) of the UE, comprising the initial eight-digit portion of the 15-digit IMEI and 16-digit IMEISV codes. See clause 6.2 of 3GPP TS 23.003 [26].<br><br>Pattern: '^[0-9]{8}$' |
| SerialNumber | string | Serial number of the UE, comprising the six-digit portion of the 15-digit IMEI and 16-digit IMEISV codes. See clause 6.2 of 3GPP TS 23.003 [26]. Leading 0s may be excluded.<br><br>Pattern: '^[0-9]{1,6}$' |

### C.3.1.4 Error Handling

General error responses are defined in clause C.1.3.

### C.3.1.5 CDDL Specification

#### C.3.1.5.1 Introduction

The data model described in clause C.3.1.3 shall be binary encoded in the CBOR format as described in IETF RFC 8949 [17].

Clause C.3.1.5.2 uses the Concise Data Definition Language described in IETF RFC 8610 [18] and provides corresponding representation of the SU_UeConfig API data model.

#### C.3.1.5.2 CDDL document

```
;;; UeConfigDoc
;;+ Represents UE configuration information associated with a VAL service.

UeConfigDoc = {
 ? UeConfigDocId: tstr
 ? configName: tstr             ; Name of the config
 valServiceDomain: tstr
 ? valServiceId: tstr
 ? valUeIds: ValUeIds
 ? ueConfigs: [+ UeConfig]
 * tstr => any
}

;;; UeConfig
;;+ UE configuration.

UeConfig = {
 configType: ConfigType
 configData: tstr               ; Actual UE configuration  data.
 * tstr => any
}

;;; ConfigType
;;+ Indicates the type of the UE configuration.

ConfigType = "COMMON" / "ON_NETWORK" / tstr ; tstr value provides forward-compatibility with future
extensions to the enumeration but is not used to encode content defined in the present version of
this API.

;;; ValUeIds
```

```
;;+ VAL UE identities for which the UE configuration is applicable.

ValUeIds = {
 ? uris: [+ Uri]
 ? imeiRanges: [+ ImeiRange]
 * tstr => any
}

;;; ImeiRange
;;+ Defines a range of IMEIs.

ImeiRange = {
 tac: TypeAllocationCode
 ? snrs: [+ SerialNumber]
 ? snrRange: SnrRange
 * tstr => any
}

;;; SnrRange
;;+ Defines a range of SerialNumbers.

SnrRange = {
 low: SerialNumber
 high: SerialNumber
}

;;; TypeAllocationCode
;;+ Type Allocation Code.

TypeAllocationCode = tstr .regexp "[0-9]{8}"

;;; SerialNumber
;;+ Serial Number.

SerialNumber = tstr .regexp "[0-9]{1,6}" ;

;;; Uri
;;+ URI

Uri = tstr          ; formatted according to RFC 3986
```

## C.3.1.6 Media Type

The media type for a user profile document shall be "application/vnd.3gpp.seal-ue-config-info+cbor".

## C.3.1.7 Media Type registration for application/vnd.3gpp.seal-ue-config-info+cbor

Type name: application

Subtype name: vnd.3gpp.seal-ue-config-info+cbor

Required parameters: none

Optional parameters: none

Encoding considerations: Must be encoded as using IETF RFC 8949 [17]. See 3GPP TS 24.546 clause C.3.1.3 for details.

Security considerations: See Section 10 of IETF RFC 8949 [17] and Section 11 of IETF RFC 7252 [12].

Interoperability considerations: Applications must ignore any key-value pairs that they do not understand. This allows backwards-compatible extensions to this specification.

Published specification: 3GPP TS 24.546 "Configuration management - Service Enabler Architecture Layer for Verticals (SEAL); Protocol specification", available via http://www.3gpp.org/specs/numbering.htm.

Applications that use this media type: Applications supporting the SEAL configuration management procedures as described in the published specification.

Fragment identifier considerations: Fragment identification is the same as specified for "application/cbor" media type in IETF RFC 8949 [17]. Note that currently that RFC does not define fragmentation identification syntax for "application/cbor".

Additional information:

Deprecated alias names for this type: N/A

Magic number(s): N/A

File extension(s): none

Macintosh file type code(s): none

Person & email address to contact for further information: <MCC name>, <MCC email address>

Intended usage: COMMON

Restrictions on usage: None

Author: 3GPP CT1 Working Group/3GPP_TSG_CT_WG1@LIST.ETSI.ORG

Change controller: <MCC name>/<MCC email address>

# Annex D (informative): Change history

| Change history | | | | | | | |
|---|---|---|---|---|---|---|---|
| Date | Meeting | TDoc | CR | Rev | Cat | Subject/Comment | New version |
| 2019-09 | CT1#120 | C1-196120 | | | | Draft skeleton provided by the rapporteur. | 0.0.0 |
| 2019-10 | CT1#120 | | | | | Implementing the following p-CR agreed by CT1: C1-196607, C1-196609, C1-196853, C1-196854 | 0.1.0 |
| 2019-11 | CT1#121 | | | | | Implementing the following p-CR agreed by CT1: C1-198620, C1-198815, C1-198816 | 0.2.0 |
| 2019-12 | CT-86 | CP-193153 | | | | Presentation for information at TSG CT | 1.0.0 |
| 2020-02 | CT1#122-e | | | | | Implementing the following p-CR agreed by CT1: C1-200645, C1-200646, C1-200873, C1-200872, C1-200649, C1-201005, C1-200823 | 1.1.0 |
| 2020-03 | CT-87e | CP-200170 | | | | Presentation for approval at TSG CT | 2.0.0 |
| 2020-03 | CT-87e | | | | | Version 16.0.0 created after approval | 16.0.0 |
| 2020-06 | CT-88e | CP-201129 | 0001 | | B | SIP based subscribe/notify procedures for configuration management | 16.1.0 |
| 2020-06 | CT-88e | CP-201129 | 0002 | 1 | F | Removal of Editor's notes. | 16.1.0 |
| 2020-06 | CT-88e | CP-201129 | 0003 | | F | Corrections in HTTP request-uri value | 16.1.0 |
| 2020-06 | CT-88e | CP-201129 | 0004 | | B | Adding IANA registration template for VAL user profile and UE configuration document | 16.1.0 |
| 2020-06 | CT-88e | CP-201129 | 0005 | | F | Using proper element names in VAL UE Configuration | 16.1.0 |
| 2020-09 | CT-89e | CP-202163 | 0006 | | D | Removing Heading level-7 as per drafting rules | 16.2.0 |
| 2021-09 | CT-93e | CP-212138 | 0007 | 1 | C | add VAL UE Information to configuration management procedure | 17.0.0 |
| 2021-12 | CT-94e | CP-213052 | 0008 | 1 | B | Addition of functional entity requirements for CoAP support | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0009 | - | B | Authenticated identity in CoAP request | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0010 | 1 | B | Addition of CoAP event subscription procedures | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0011 | 1 | B | Addition of CoAP notifications procedure | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0012 | 1 | B | Addition of CoAP VAL user profile data procedures | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0013 | 2 | B | Addition of CoAP Update VAL user profile data procedures | 17.1.0 |
| 2021-12 | CT-94e | CP-213052 | 0014 | 1 | B | Addition of CoAP resource representation and encoding | 17.1.0 |
| 2022-03 | CT-95e | CP-220255 | 0015 | 2 | B | Data types applicable to multiple resource representations | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0016 | 2 | B | Addition of CoAP VAL UE configuration data procedures | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0017 | - | F | Minor corrections in VAL user profile data procedures | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0018 | - | B | Media type for user profile document | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0019 | - | B | Resolving Editor's Note on CoAP use of cache | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0020 | 2 | F | Corrections in CoAP Resource representation and APIs for VAL user profile | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0021 | 3 | B | Addition of CoAP Resource representation and APIs for UE configuration | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0022 | - | F | Correction of CR implementation issues | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0023 | - | B | Corrections in Update VAL user profile data procedures | 17.2.0 |
| 2022-03 | CT-95e | CP-220255 | 0024 | - | B | Updates in VAL user profile data SCM server CoAP procedure | 17.2.0 |
| 2022-06 | CT-96 | CP-221217 | 0025 | - | D | Correction on Annex numbers referred in VAL UE configuration data | 17.3.0 |
| 2022-06 | CT-96 | CP-221217 | 0026 | - | F | Resolve editor's notes | 17.3.0 |
| 2022-06 | CT-96 | CP-221217 | 0027 | - | F | CoAP procedure alignments | 17.3.0 |
| 2022-06 | CT-96 | CP-221217 | 0028 | - | F | Reference update | 17.3.0 |
| 2022-06 | CT-96 | CP-221217 | 0029 | - | F | Updates to error handling | 17.3.0 |
| 2022-06 | CT-96 | CP-221217 | 0030 | - | F | Updates to data types | 17.3.0 |
| 2022-09 | CT-97e | CP-222150 | 0031 | 1 | B | Update of resource representation and encoding annex | 17.4.0 |
| 2022-12 | CT-98e | CP-223123 | 0033 | | F | Reference update | 17.5.0 |
| 2023-03 | CT-99 | CP-230233 | 0037 | - | F | Alignment with CDDL specification and miscellaneous corrections | 17.6.0 |
| 2023-03 | CT-99 | CP-230233 | 0036 | 1 | F | Reference update: RFC 9177 | 17.6.0 |
| 2023-03 | CT-99 | CP-230248 | 0035 | 1 | A | Correction to the XML schema | 17.6.0 |
| 2023-03 | CT-99 | CP-230220 | 0038 | - | F | Miscellaneous corrections | 18.0.0 |
| 2025-03 | CT#107 | CP-250151 | 0042 | 1 | A | Resolution of editor's note under clause C.3.1.6 | 18.1.0 |
| 2025-03 | CT#107 | CP-250151 | 0044 | 1 | A | Resolution of editor's note under clause C.3.1.2.1 | 18.1.0 |
| 2025-06 | CT#108 | CP-251193 | 0048 | 2 | A | Correction to the XML schema on element names | 18.2.0 |
| 2025-12 | CT#110 | CP-253191 | 0061 | | A | Fixing the SU_UeConfig API- CDDL | 18.3.0 |
| 2025-12 | CT#110 | CP-253191 | 0064 | | A | Fixing the SU_UserProfile API - CDDL | 18.3.0 |
| 2026-09 | CT#113 | CP-262057 | 0079 | 1 | A | Resolution of editor's note under clause C.2.1 | 18.4.0 |

# History

| Version | Date | Status |
|---|---|---|
| V18.0.0 | May 2024 | Publication |
| V18.1.0 | March 2025 | Publication |
| V18.2.0 | July 2025 | Publication |
| V18.3.0 | February 2026 | Publication |
| V18.4.0 | September 2026 | Publication |